

that there is a quadratic polynomial  $P(z)$  such that

$$u(z) = \operatorname{Re}(P(z)) + \frac{(\Delta u)(z_0)}{4}|z - z_0|^2 + R(z),$$

where  $R(z) = o(|z - z_0|^2)$ , that is,  $R(z)/|z - z_0|^2 \rightarrow 0$  as  $z \rightarrow z_0$ .

(b) Use (a) to show that if  $(\Delta u)(z_0) > 0$ , then  $u(z)$  satisfies the mean value inequality at  $z_0$ . (c) Use (a) and (b) to show that  $u(x, y)$  is subharmonic on  $D$  if and only if  $(\Delta u)(z) \geq 0$  on  $D$ . *Hint.* If  $\Delta u \geq 0$ , show first that  $u + \varepsilon|z|^2$  is subharmonic for any  $\varepsilon > 0$ .

6. Show that if  $u(x, y)$  is harmonic and  $p > 0$ , then  $|\nabla u|^p$  is subharmonic. *Hint.* Use Exercise 5c.
7. Let  $h(z)$  be a smooth real-valued function on a neighborhood of the closure of a bounded domain  $D$ . Show that  $h(z)$  can be expressed as the difference of two smooth subharmonic functions. *Hint.* Consider  $u(z) = h(z) + C|z|^2$ .
8. Let  $E$  be a subset of  $\partial D$  for which there is a subharmonic function  $v(z)$  on  $D$  such that  $v(z) < 0$  for  $z \in D$ , and  $v(z)$  tends to  $-\infty$  as  $z \in D$  tends to any point of  $E$ . Show that if  $u(z)$  is a subharmonic function on  $D$  that is bounded above, and if  $\limsup u(z) \leq 0$  as  $z \in D$  tends to any point of  $\partial D$  that is not in  $E$ , then  $u(z) \leq 0$  in  $D$ .
9. A real-valued function  $\chi(t)$  on an interval  $I \subseteq \mathbb{R}$  is **convex** if  $\chi(st_0 + (1-s)t_1) \leq s\chi(t_0) + (1-s)\chi(t_1)$  whenever  $t_0, t_1 \in I$  and  $0 \leq s \leq 1$ .
  - (a) Show that a convex function on an open interval is continuous.
  - (b) Show that a smooth function  $\chi(t)$  is convex if and only if  $\chi''(t) \geq 0$ .
  - (c) Show that if  $\chi(t)$  is convex, and if  $s_1, \dots, s_n \geq 0$  satisfy  $\sum s_j = 1$ , then

$$\chi\left(\sum_{j=1}^n s_j t_j\right) \leq \sum_{j=1}^n s_j \chi(t_j), \quad t_1, \dots, t_n \in I.$$

(d) Suppose  $w(t)$  is a continuous function on an interval  $J$  such that  $w(t) \geq 0$  and  $\int_J w(t)dt = 1$ . Show that if  $\chi(t)$  is a convex function on  $I$ , and if  $h(t)$  is any continuous function from  $J$  to  $I$ , then

$$\chi\left(\int_J h(t)w(t)dt\right) \leq \int_J \chi(h(t))w(t)dt.$$

*Remark.* This is **Jensen's inequality**. The relevant property of  $w(t)$  is that  $w(t)dt$  is a probability measure. To prove Jensen's inequality, approximate the integral by sums and use (c).

10. Show that if  $u(z)$  is a subharmonic function on a domain  $D$  with values in an interval  $I$ , and if  $\chi(t)$  is an increasing convex function on  $I$ , then  $\chi(u(z))$  is subharmonic. *Hint.* Use Jensen's inequality.
11. Show that if  $f(z)$  is analytic and  $p > 0$ , then  $|f(z)|^p$  is subharmonic. *Hint.* Use the convexity of  $e^{pt}$  and Exercise 10, or use Exercise 6.

### 3. Compactness of Families of Harmonic Functions

The Arzelà-Ascoli theorem asserts that a uniformly bounded family of equicontinuous functions is compact. For a family of analytic functions, the Cauchy integral representation formula shows that uniform boundedness already implies equicontinuity. Hence a uniformly bounded family of analytic functions is compact. We can draw the same conclusion for a uniformly bounded family of harmonic functions using the Poisson integral representation formula. The Poisson integral formula for the unit disk is

$$(3.1) \quad u(re^{i\theta}) = \int_0^{2\pi} u(e^{i\varphi}) P_r(\theta - \varphi) \frac{d\varphi}{2\pi} = \int_0^{2\pi} u(e^{i(\theta - \varphi)}) P_r(\varphi) \frac{d\varphi}{2\pi},$$

where

$$(3.2) \quad P_r(\theta) = \frac{1 - r^2}{1 + r^2 - 2r \cos \theta}, \quad 0 \leq r < 1, \quad 0 \leq \theta \leq 2\pi.$$

The Poisson kernel is an infinitely differentiable function of  $x$  and  $y$ . We can differentiate (3.1) under the integral sign with respect to  $x$  and  $y$ , and when we do, we obtain integral representations of the derivatives of  $u(x, y)$  in terms of the function  $u(e^{i\theta})$  and the derivatives of the Poisson kernel,

$$\frac{\partial}{\partial x} u(z) = \int_0^{2\pi} u(e^{i\varphi}) \frac{\partial}{\partial x} P_r(\theta - \varphi) \frac{d\varphi}{2\pi}, \quad z = x + iy = re^{i\theta} \in \mathbb{D},$$

with similar formulae for the other derivatives. These integral formulae allow us to estimate the derivatives of  $u(x, y)$  in terms of the maximum of  $M$  of  $|u(e^{i\theta})|$  on the unit circle,

$$\left| \frac{\partial}{\partial x} u(z) \right| \leq M \int_0^{2\pi} \left| \frac{\partial}{\partial x} P_r(\theta - \varphi) \right| \frac{d\varphi}{2\pi}, \quad z = x + iy = re^{i\theta} \in \mathbb{D},$$

and similarly for the other derivatives. Since each derivative of the Poisson kernel is uniformly bounded on each proper subdisk of the unit disk, we obtain uniform bounds for each derivative of  $u(z)$  in terms of the maximum of  $|u(z)|$  on the unit circle. (See also Section X.1, Exercises 8 and 9.)

The same procedure works if we replace the Poisson integral representation for the unit disk by the Poisson integral representation for an arbitrary closed disk. We obtain thus the following.

**Lemma.** *If  $\mathcal{F}$  is a family of harmonic functions on a domain  $D$  that is uniformly bounded on each closed disk in  $D$ , then the partial derivatives of the functions in  $\mathcal{F}$  of any fixed order are uniformly bounded on each closed disk in  $D$ .*

In particular, the gradients of the functions in  $\mathcal{F}$  are uniformly bounded on any closed disk in  $D$ , so the functions in  $\mathcal{F}$  are equicontinuous. (See Section XI.2.) From the Arzelà-Ascoli theorem and a diagonalization argument, we obtain the following analogue for harmonic functions of Montel's theorem.

**Theorem.** *Let  $\mathcal{F}$  be a family of harmonic functions on a domain  $D$  that is uniformly bounded on each compact subset of  $D$ . Then every sequence in  $\mathcal{F}$  has a subsequence that converges uniformly on each compact subset of  $D$ .*

Families of positive harmonic functions are also compact, provided that we allow  $+\infty$  as a possible limit for a sequence of positive harmonic functions. This follows again from the Poisson integral representation. This time it is the positivity of the Poisson kernel that is crucial.

**Theorem (Harnack's Inequality).** *If  $u(z)$  is a positive harmonic function on the open unit disk  $\mathbb{D}$ , then*

$$(3.3) \quad \frac{1-r}{1+r} u(0) \leq u(re^{i\theta}) \leq \frac{1+r}{1-r} u(0), \quad re^{i\theta} \in \mathbb{D}.$$

The proof depends on the estimate

$$\frac{1-r}{1+r} = \frac{1-r^2}{1+r^2+2r} \leq P_r(\theta) \leq \frac{1-r^2}{1+r^2-2r} = \frac{1+r}{1-r}.$$

We can always approximate  $u(z)$  by a dilate  $u(\rho z)$ ,  $\rho < 1$ , and assume that  $u(z)$  extends harmonically across the unit circle. Then  $u(z)$  is represented by the Poisson integral formula (3.1). Substituting the estimates for  $P_r(\theta)$  into (3.1) and using the positivity of  $u(z)$ , we obtain

$$\begin{aligned} \frac{1-r}{1+r} \int_0^{2\pi} u(e^{i(\theta-\varphi)}) \frac{d\varphi}{2\pi} &\leq \int_0^{2\pi} u(e^{i(\theta-\varphi)}) P_r(\varphi) \frac{d\varphi}{2\pi} \\ &\leq \frac{1+r}{1-r} \int_0^{2\pi} u(e^{i(\theta-\varphi)}) \frac{d\varphi}{2\pi}. \end{aligned}$$

Since the average value of  $u(e^{i(\theta-\varphi)})$  is  $u(0)$ , this becomes (3.3).

The estimate (3.3) is sharp. It becomes an equality for the function  $u(z) = \operatorname{Re}((1+z)/(1-z))$  at  $z = \pm r$ .

If we scale and translate Harnack's inequality to an arbitrary disk, of radius  $R > 0$  and center  $z_0$ , we obtain

$$(3.4) \quad \frac{R-r}{R+r} u(z_0) \leq u(z) \leq \frac{R+r}{R-r} u(z_0), \quad |z - z_0| \leq r, \quad r < R,$$

for all positive harmonic functions  $u(z)$  on the disk  $|z - z_0| < R$ . The estimate (3.4) is also referred to as **Harnack's inequality**. It allows us to draw some striking conclusions. From the left-hand inequality it follows that if  $\{u_n(z)\}$  is a sequence of positive harmonic functions on the disk  $\{|z - z_0| < R\}$  such that  $u_n(z_0) \rightarrow +\infty$ , then  $u_n(z) \rightarrow +\infty$  uniformly on any subdisk  $\{|z - z_0| \leq \rho\}$ ,  $\rho < R$ . From the right-hand inequality it follows that if  $\{u_n(z_0)\}$  is bounded, then  $u_n(z)$  is uniformly bounded on any subdisk  $\{|z - z_0| \leq \rho\}$ ,  $\rho < R$ . If we combine these observations with a connectedness argument, we obtain the following.

**Lemma.** *Suppose  $\{u_n(z)\}$  is a sequence of positive harmonic functions on a domain  $D$ . If there is  $z_0 \in D$  such that  $u_n(z_0) \rightarrow +\infty$ , then  $u_n(z) \rightarrow +\infty$  uniformly on each compact subset of  $D$ . If there is  $z_0 \in D$  such that  $\{u_n(z_0)\}$  is bounded, then  $\{u_n(z)\}$  is uniformly bounded on each compact subset of  $D$ .*

To see this, let  $U$  be the set of  $z \in D$  such that  $u_n(z) \rightarrow +\infty$ . The remarks preceding the lemma show that both  $U$  and  $D \setminus U$  are open. Since  $D$  is a domain, either  $U = D$  or  $U$  is empty. In the case  $U = D$ , we can cover any compact set by a finite number of disks on which  $u_n(z) \rightarrow +\infty$  uniformly, and we see that  $u_n(z) \rightarrow +\infty$  uniformly on compact subsets of  $D$ . If  $U$  is empty, we cover any compact set by a finite number of disks, and we see that  $\{u_n(z)\}$  is uniformly bounded on each compact subset of  $D$ .

If we combine the preceding lemma and the compactness theorem for families of harmonic functions, we obtain immediately the following compactness theorem for families of positive harmonic functions.

**Theorem.** *Let  $\mathcal{F}$  be a family of positive harmonic functions on a domain  $D$ . Every sequence in  $\mathcal{F}$  has a subsequence that either converges uniformly on compact subsets of  $D$  to a harmonic function or converges uniformly on compact subsets of  $D$  to  $+\infty$ .*

The compactness theorem for positive harmonic functions can be applied to monotone sequences of harmonic functions.

**Theorem.** *Let  $\{u_n(z)\}$  be an increasing sequence of harmonic functions on a domain  $D$ . Then  $\{u_n(z)\}$  converges uniformly on compact subsets of  $D$ , either to a harmonic function or to  $+\infty$ .*

To see this, consider the functions  $v_n(z) = u_n(z) - u_1(z)$ . For each fixed  $z$  the sequence  $\{v_n(z)\}$  is increasing, hence convergent either to a finite value or to  $+\infty$ . The preceding theorem shows that  $\{v_n(z)\}$  has a subsequence that converges uniformly on compact subsets of  $D$ , either to a harmonic function or to  $+\infty$ . It follows that  $\{u_n(z)\}$  also converges uniformly on compact subsets of  $D$ .

### Exercises for XV.3

1. Let  $\{u_n(z)\}$  be a sequence of positive harmonic functions on a domain  $D$ . Show that either  $\sum u_n(z)$  converges to a harmonic function uniformly on each compact subset of  $D$ , or the partial sums of the series converge uniformly to  $+\infty$  on each compact subset of  $D$ .
2. Let  $u(z)$  be a positive harmonic function on the horizontal strip  $\{-1 < \operatorname{Im} z < 1\}$ . Show that if  $u(n) \rightarrow +\infty$  as  $n \rightarrow +\infty$ , then  $u(x + iy) \rightarrow +\infty$  as  $x \rightarrow +\infty$  for all  $y$ ,  $-1 < y < 1$ .
3. Let  $u(z)$  be a positive harmonic function on the horizontal strip  $\{-1 < \operatorname{Im} z < 1\}$ . Show that if  $u(n) \rightarrow 0$  as  $n \rightarrow +\infty$ , then  $u(x + iy) \rightarrow 0$  as  $x \rightarrow +\infty$  for all  $y$ ,  $-1 < y < 1$ .
4. Let  $K$  be a compact subset of a domain  $D$ . Show that there are constants  $c > 0$  and  $C > 1$  such that  $c \leq u(z)/u(w) \leq C$  for all positive harmonic functions  $u$  on  $D$  and all points  $z, w \in K$ .
5. Let  $D$  be a bounded domain. For  $z, w \in D$ , define  $d(z, w)$  to be the infimum of  $\log C$ , taken over all constants  $C > 1$  such that  $(1/C) \leq u(z)/u(w) \leq C$  for all positive harmonic functions  $u$  on  $D$ . (a) Show that  $d(z, w) \leq d(z, \zeta) + d(\zeta, w)$  for all  $z, \zeta, w \in D$ . (b) Show that  $d(z_n, z_0) \rightarrow 0$  if and only if  $|z_n - z_0| \rightarrow 0$ . (c) Find  $d(z, w)$  in the case that  $D = \mathbb{D}$  is the open unit disk.
6. Prove the compactness theorem for families of positive harmonic functions by applying Montel's theorem to functions of the form  $f = e^{-(u+iv)}$ , where  $v$  is a (locally defined) harmonic conjugate for the positive harmonic function  $u$ .
7. Let  $u(z)$  be a harmonic function on a domain  $D$ . Show that if  $|u(z)| \leq M$  for  $z \in D$ , then  $|(\nabla u)(z)| \leq 2M/\operatorname{dist}(z, \partial D)$ . *Hint.* Use "scaling." Check the estimate for the unit disk and  $z = 0$  from the Poisson integral representation, and transfer the estimate to a disk centered at  $z$  of appropriate radius. The gradient scales according to one over the radius of the disk.

## 4. The Perron Method

Let  $D$  be a bounded domain. Recall that the Dirichlet problem for a given continuous function  $h(z)$  on the boundary of  $D$  is to find a harmonic function  $u(z)$  on  $D$  that has boundary values  $h(z)$ . Any such function  $u(z)$  is unique. However, the Dirichlet problem need not be solvable.

**Example.** Let  $D = \mathbb{D} \setminus \{0\}$  be the punctured unit disk, and let  $h(z)$  be the continuous function on  $\partial D$  defined by  $h(z) = 0$  for  $|z| = 1$ , while  $h(0) = 1$ . If  $u(z)$  is a harmonic function on  $D$  such that  $\limsup u(z) \leq h(\zeta)$  for  $\zeta \in \partial D$ , then for any  $\varepsilon > 0$ ,  $u(z) \leq \varepsilon \log(1/|z|)$  for  $|z| = 1$  and also for  $z$  near 0. By the maximum principle, the estimate holds on all of  $D$ . Letting  $\varepsilon \rightarrow 0$ , we obtain  $u(z) \leq 0$  on  $D$ . Consequently,  $u(z)$  cannot assume the boundary value 1 at  $z = 0$ .

Let  $h(\zeta)$  be a continuous real-valued function on  $\partial D$ . We define the **Perron family of subsolutions** corresponding to  $h(\zeta)$ , denoted by  $\mathcal{F}_h$ , to be the family of all subharmonic functions  $u(z)$  on  $D$  such that

$$\limsup_{D \ni z \rightarrow \zeta} u(z) \leq h(\zeta), \quad \zeta \in \partial D.$$

We define the **Perron solution** to the Dirichlet problem corresponding to the boundary function  $h(\zeta)$  to be the upper envelope  $\tilde{h}(z)$  of the family  $\mathcal{F}_h$ ,

$$\tilde{h}(z) = \sup\{u(z) : u \in \mathcal{F}_h\}, \quad z \in D.$$

In the case that there is a harmonic function  $u(z)$  on  $D$  that attains the boundary values  $h(\zeta)$  continuously on  $\partial D$ , then happily  $u(z)$  coincides with the Perron solution. Indeed, since  $u \in \mathcal{F}_h$ , we have  $u(z) \leq \tilde{h}(z)$  on  $D$ . The reverse inequality follows from the maximum principle. If  $v \in \mathcal{F}_h$ , then by the maximum principle,  $v(z) \leq u(z)$  on  $D$ , and consequently,  $\tilde{h}(z) \leq u(z)$  on  $D$ .

Several elementary properties of the Perron solution are listed in the next lemma.

**Lemma.** *Let  $D$  be a bounded domain, and let  $h(\zeta)$  be a continuous real-valued function on  $\partial D$ .*

- (i) *If  $\alpha \leq h(\zeta) \leq \beta$  on  $\partial D$ , then  $\alpha \leq \tilde{h}(z) \leq \beta$  on  $D$ .*
- (ii) *If  $\widetilde{h_1}(\zeta) \leq \widetilde{h_2}(\zeta)$  on  $\partial D$ , then  $\widetilde{h_1}(z) \leq \widetilde{h_2}(z)$  on  $D$ .*
- (iii)  *$a\tilde{h}(z) = \widetilde{ah}(z)$  for all  $a > 0$ .*
- (iv)  *$\widetilde{h_1}(z) + \widetilde{h_2}(z) \leq \widetilde{(h_1 + h_2)}(z)$ ,  $z \in D$ .*
- (v)  *$\widetilde{(h + v)}(z) = \tilde{h}(z) + v(z)$  on  $D$  whenever  $v(z)$  is continuous on  $D \cup \partial D$  and harmonic on  $D$ .*

In (i), the constant function  $\alpha$  belongs to  $\mathcal{F}_h$ , so  $\alpha \leq \tilde{h}$ . The estimate  $\tilde{h} \leq \beta$  follows from the maximum principle, as above. Properties (ii), (iii),

and (v) follow virtually immediately from the definitions. For (ii) note that  $\mathcal{F}_{h_1} \subseteq \mathcal{F}_{h_2}$ , for (iii) that  $\mathcal{F}_{ah} = a\mathcal{F}_h$ , and for (v) that  $\mathcal{F}_{h+v} = \mathcal{F}_h + v$ . To prove (iv), suppose  $u_1 \in \mathcal{F}_{h_1}$  and  $u_2 \in \mathcal{F}_{h_2}$ . Then  $u_1 + u_2 \in \mathcal{F}_{h_1+h_2}$ , and consequently,  $u_1(z) + u_2(z) \leq \widetilde{(h_1 + h_2)}(z)$ . If we take the supremum over  $u_1 \in \mathcal{F}_{h_1}$  and  $u_2 \in \mathcal{F}_{h_2}$ , we obtain (iv).

**Theorem.** *The Perron solution  $\tilde{h}(z)$  defined above is harmonic on  $D$ .*

For the proof, let  $D_0$  be a fixed disk such that  $D_0 \cup \partial D_0 \subset D$ , and let  $\{z_j\}_{j=1}^\infty$  be a sequence of points in  $D_0$  that is dense in  $D_0$ . Let  $\{u_{jk}(z)\}_{k=1}^\infty$  be a sequence of functions in  $\mathcal{F}_h$  such that  $u_{jk}(z_j) \rightarrow \tilde{h}(z_j)$  for  $j \geq 1$ . Define  $v_m(z) = \max\{u_{jk}(z) : 1 \leq j, k \leq m\}$ . Then  $v_m \in \mathcal{F}_h$ ,  $\{v_m(z)\}$  is an increasing sequence, and  $v_m(z_j) \rightarrow \tilde{h}(z_j)$  for all  $j$ . Define  $w_m(z)$  so that  $w_m(z) = v_m(z)$  for  $z \in D \setminus D_0$ , and  $w_m(z)$  is the harmonic extension of  $v_m|_{\partial D_0}$  on  $D_0$ . Then  $w_m \in \mathcal{F}_h$ ,  $\{w_m(z)\}$  is an increasing sequence, and  $w_m(z_j) \rightarrow \tilde{h}(z_j)$  for all  $j$ . Further, the functions  $w_m(z)$  are all harmonic on  $D_0$ . By Harnack's theorem, the sequence  $\{w_m(z)\}$  converges uniformly on compact subsets of  $D_0$  to a harmonic function  $w(z)$ , which evidently satisfies  $w(z_j) = \tilde{h}(z_j)$  for all  $j$ . Now let  $z_0$  be any other point of  $D_0$ . We go through this procedure for the sequence  $\{z_j\}_{j=0}^\infty$  obtained by adjoining  $z_0$  to the original sequence, and we obtain a harmonic function  $w_0(z)$  on  $D_0$  such that  $w_0(z_j) = \tilde{h}(z_j)$  for all  $j \geq 0$ . Since  $w(z) = w_0(z)$  on a dense subset of  $D_0$ , in fact  $w(z) = w_0(z)$  for all  $z \in D_0$ , and in particular  $w(z_0) = w_0(z_0) = \tilde{h}(z_0)$ . Since  $z_0$  is arbitrary,  $w(z) = \tilde{h}(z)$  for all  $z \in D_0$ , and  $\tilde{h}(z)$  is harmonic on  $D_0$ , hence on  $D$ .

Now we address the problem of whether  $\tilde{h}(z)$  has boundary values  $h(\zeta)$  on  $\partial D$ . We say that a point  $\zeta_0 \in \partial D$  is a **regular boundary point of  $D$**  if

$$\lim_{D \ni z \rightarrow \zeta_0} \tilde{h}(z) = h(\zeta_0)$$

for any continuous function  $h(\zeta)$  on  $\partial D$ .

**Example.** The origin  $\zeta_0 = 0$  is *not* a regular boundary point of the punctured unit disk  $D = \mathbb{D} \setminus \{0\}$ . We will see shortly that each boundary point on the unit circle is regular.

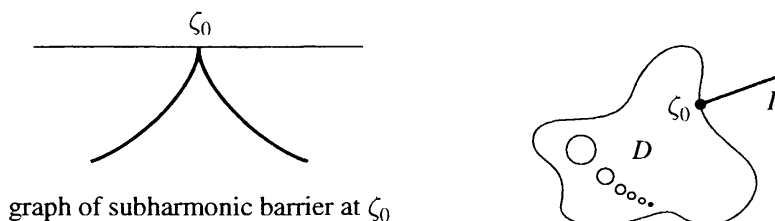
Suppose that  $\zeta_0 \in \partial D$  is a regular boundary point. We solve the Dirichlet problem for the function  $h(\zeta) = |\zeta - \zeta_0|$ ,  $\zeta \in \partial D$ . Since the function  $|z - \zeta_0|$  belongs to  $\mathcal{F}_h$ , the Perron solution  $\tilde{h}(z)$  satisfies  $\tilde{h}(z) \geq |z - \zeta_0|$ . Since  $\zeta_0$  is regular,  $\tilde{h}(z) \rightarrow 0$  as  $z \rightarrow \zeta_0$ . Thus the function  $w(z) = -\tilde{h}(z)$  is harmonic on  $D$ ,  $w(z) \leq -|z - \zeta_0|$  on  $D$ , and  $w(z) \rightarrow 0$  as  $z \rightarrow \zeta_0$ . The function  $w(z)$  “peaks” at  $\zeta_0$ , in the sense that  $w(z)$  has a limit at  $\zeta_0$  that is strictly greater than  $\limsup w(z)$  as  $z$  tends to any other boundary point of  $D$ . This motivates the following definition.

A **subharmonic barrier** at  $\zeta_0 \in \partial D$  is a subharmonic function  $w(z)$  defined on  $D \cap \{|z - \zeta_0| < \delta\}$  for some  $\delta > 0$ , such that

- (i)  $w(z) < 0$  for  $z \in D$ ,  $|z - \zeta_0| < \delta$ ,
- (ii)  $w(z) \rightarrow 0$  as  $z \in D$  tends to  $\zeta_0$ ,
- (iii)  $\limsup w(z) < 0$  as  $z \in D$  tends to  $\zeta$  for any  $\zeta \in \partial D$ ,  $0 < |\zeta - \zeta_0| < \delta$ .

While only defined on the part of  $D$  near  $\zeta_0$ , the subharmonic barrier can be adjusted so that it is defined on all of  $D$  and satisfies (iii) at any  $\zeta \in \partial D \setminus \{\zeta_0\}$ . The conditions (i) and (iii) guarantee that if  $\varepsilon > 0$  is sufficiently small, then  $w(z) \leq -2\varepsilon$  for all  $z \in D$  that satisfy  $|z - \zeta_0| = \delta/2$ . If we set  $u(z) = \max(w(z), -\varepsilon)$  for  $z \in D \cap \{|z - \zeta_0| < \delta/2\}$ , and  $u(z) = -\varepsilon$  for other points of  $D$ , then  $u(z)$  is a subharmonic barrier at  $\zeta_0$  that is now defined on all of  $D$ .

**Example.** In the case  $D = \mathbb{D}$  is the open unit disk, the function  $w(z) = \operatorname{Re}(\bar{\zeta}_0 z) - 1$  is a subharmonic barrier at the boundary point  $\zeta_0 \in \partial \mathbb{D}$ . In this case, the barrier is harmonic.



**Example.** Suppose there is a straight line segment  $I$  in  $\mathbb{C} \setminus D$  that has one endpoint at  $\zeta_0 \in \partial D$ . We map the complement  $\mathbb{C}^* \setminus I$  of the segment onto the open unit disk  $\mathbb{D}$  by a conformal map  $\varphi(z)$  such that  $\varphi(\zeta_0) = 1$ . Then  $w(z) = \operatorname{Re} \varphi(z) - 1$  is a subharmonic barrier at  $\zeta_0$ .

**Example.** If  $D$  is bounded by a finite number of smooth boundary curves, then every point of  $\partial D$  has a subharmonic barrier. In this case, each  $\zeta_0 \in \partial D$  is the endpoint of a line segment contained in  $\mathbb{C} \setminus D$ , in the direction of the outer normal to  $D$  at  $\zeta_0$ .

We showed above that if  $\zeta_0$  is a regular boundary point of  $D$ , then there is a subharmonic barrier at  $\zeta_0$ . The following theorem shows that the existence of subharmonic barriers characterizes regular boundary points.

**Theorem.** Let  $D$  be a bounded domain, and let  $\zeta_0 \in \partial D$ . If there is a subharmonic barrier at  $\zeta_0$ , then  $\zeta_0$  is a regular boundary point of  $D$ .

To prove this, let  $h(\zeta)$  be a continuous real-valued function on  $\partial D$ . We assume for convenience that  $h(\zeta_0) = 0$  and that  $|h(\zeta)| \leq 1$  for  $\zeta \in \partial D$ . Let  $\varepsilon > 0$ , and choose  $\delta > 0$  such that  $|h(\zeta)| < \varepsilon$  for  $\zeta \in \partial D$ ,  $|\zeta - \zeta_0| \leq \delta$ . Let  $w(z)$  be a subharmonic barrier at  $\zeta_0$ . We can assume that the subharmonic



barrier  $w(z)$  is defined on all of  $D$ . Choose  $\rho > 0$  such that  $w(z) \leq -\rho$  for  $|z| \geq \delta$ . Then  $u(z) = (w(z)/\rho) - \varepsilon$  is subharmonic on  $D$ ,  $u(z) \leq -\varepsilon$  on  $D$ , and  $\limsup_{z \rightarrow \zeta} u(z) \leq -1$  for any point  $\zeta \in \partial D$  satisfying  $|\zeta| \geq \delta$ . Thus  $\limsup_{z \rightarrow \zeta} u(z) \leq h(\zeta)$  for all  $\zeta \in \partial D$ , and  $u \in \mathcal{F}_h$ . Hence  $u(z) \leq \tilde{h}(z)$ . Since  $u(z) \rightarrow -\varepsilon$  as  $z \rightarrow \zeta_0$ , we obtain  $\tilde{h}(z) \geq -2\varepsilon$  for  $\zeta$  near  $\zeta_0$ . The same argument, applied to  $-h(\zeta)$ , shows that  $\widetilde{(-h)}(z) \geq -2\varepsilon$  for  $\zeta$  near  $\zeta_0$ . By part (iv) of the lemma,  $\tilde{h}(z) + \widetilde{(-h)}(z) \leq 0$ . Hence  $\tilde{h}(z) \leq -\widetilde{(-h)}(z) \leq 2\varepsilon$  for  $\zeta$  near  $\zeta_0$ . It follows that  $|\tilde{h}(z)| \leq 2\varepsilon$  for  $\zeta$  near  $\zeta_0$ . Since this is true for any  $\varepsilon > 0$ ,  $\tilde{h}(z) \rightarrow 0$  as  $z \rightarrow \zeta_0$ .

Since there is a subharmonic barrier at each smooth boundary point, we obtain the following corollary.

**Corollary.** *Let  $D$  be a bounded domain whose boundary  $\partial D$  consists of a finite number of piecewise smooth closed curves. If  $h(\zeta)$  is a continuous function on  $\partial D$ , then  $\tilde{h}(z)$  is a harmonic function on  $D$  that tends to  $h(\zeta)$  as  $z \in D$  tends to each boundary point  $\zeta$  of  $\partial D$ .*

#### Exercises for XV.4

1. Let  $D$  be the annulus  $\{a < |z| < b\}$ . Find  $\tilde{h}(z)$  for the function  $h(\zeta)$  defined by  $h(\zeta) = \alpha$  for  $|\zeta| = a$  and  $h(\zeta) = \beta$  for  $|\zeta| = b$ .
2. Let  $D = \mathbb{D} \setminus \{0\}$  be the punctured unit disk, and let  $h(\zeta)$  be a continuous function on  $\partial D$ . Show that  $\tilde{h}(z)$  is the Poisson integral of the restriction of  $h(\zeta)$  to the unit circle  $\partial \mathbb{D}$ , irrespective of the value  $h(0)$ .
3. Let  $D$  be a bounded domain, and let  $h(\zeta)$  be continuous on  $\partial D$ . Let  $z_0 \in D$ , and for  $\varepsilon > 0$ , set  $D_\varepsilon = D \cap \{|z - z_0| > \varepsilon\}$ . Define  $h_\varepsilon$  on  $\partial D_\varepsilon$  so that  $h_\varepsilon = h$  on  $\partial D$ , and  $h_\varepsilon(\zeta) = 0$  for  $|\zeta - z_0| = \varepsilon$ . Show that  $\tilde{h}_\varepsilon(z)$  converges to  $\tilde{h}(z)$  as  $\varepsilon \rightarrow 0$ , uniformly on any subset of  $D$  at a positive distance from  $z_0$ .
4. Let  $D$  be a bounded domain, and let  $h(z)$  be a continuous function on  $D \cup \partial D$ . Let  $\{D_m\}$  be an increasing sequence of domains, each bounded by a finite number of smooth closed curves, such that  $D_m \cup \partial D_m \subset D_{m+1}$  and  $\cup D_m = D$ . Let  $u_m$  be the harmonic extension of  $h|_{\partial D_m}$  to  $D_m$ . Show that  $u_m(z)$  converges uniformly on compact subsets of  $D$  to  $\tilde{h}(z)$ . *Hint.* Suppose first that  $h(z)$  is smooth, and represent  $h(z)$  as the difference of two subharmonic functions, as in Exercise 2.6.
5. Let  $D$  be a bounded domain, and let  $h_1(\zeta)$  and  $h_2(\zeta)$  be continuous functions on  $\partial D$ . Show that  $\widetilde{(h_1 + h_2)}(z) = \tilde{h}_1(z) + \tilde{h}_2(z)$  for  $z \in D$ . *Remark.* Thus the correspondence  $h \mapsto \tilde{h}$  is linear. This follows

from the solvability of the Dirichlet problem in the case that  $\partial D$  is smooth. To prove it in the general case, use Exercise 4.

## 5. The Riemann Mapping Theorem Revisited

The Riemann mapping theory can be proved by solving a Dirichlet problem. This was Riemann's original strategy, though he could not solve the Dirichlet problem in the full generality of simply connected domains. We will give a proof along the lines envisioned by Riemann.

We say that a subset  $E$  of the extended complex plane  $\mathbb{C}^*$  is a **continuum** if it is compact and connected and has more than one point. If  $E$  is a continuum, then  $\mathbb{C}^* \setminus E$  is simply connected. Conversely, if  $E$  is a compact set such that  $\mathbb{C}^* \setminus E$  is simply connected, and if  $E$  has more than one point, then  $E$  is a continuum. (See Section VIII.8.) The following criterion allows us to solve the Dirichlet problem for any bounded simply connected domain.

**Lemma.** *Let  $D$  be a domain, and let  $\zeta_0 \in \partial D$ . If  $\zeta_0$  lies on a continuum in  $\mathbb{C} \setminus D$ , then  $\zeta_0$  is a regular boundary point of  $D$ .*

We construct a subharmonic barrier at  $\zeta_0$ . Let  $E$  be the continuum in  $\mathbb{C} \setminus D$  containing  $\zeta_0$ , and let  $\zeta_1$  be any other point of  $E$ . Since  $\mathbb{C}^* \setminus E$  is simply connected,  $\log((z - \zeta_0)/(z - \zeta_1))$  has an analytic branch on  $\mathbb{C} \setminus E$ ; call it  $f(z)$ . The set  $D_0 = \{|z - \zeta_0|/|z - \zeta_1| < 1/e\}$  is a disk containing  $\zeta_0$ , and the values of  $w = f(z)$  on  $D_0$  lie in the half-plane  $\{\operatorname{Re} w < -1\}$ . Thus the values of  $1/f(z)$  on  $D \cap D_0$  lie in the disk  $\{|w + \frac{1}{2}| < \frac{1}{2}\}$  in the left half-plane. Since  $\operatorname{Re} f(z) \rightarrow -\infty$  if and only if  $z \rightarrow \zeta_0$ ,  $1/f(z) \rightarrow 0$  for  $z \in D \cap D_0$  if and only if  $z \rightarrow \zeta_0$ . Hence the restriction of  $\operatorname{Re}(1/f(z))$  to  $D \cap D_0$  is a harmonic barrier at  $\zeta_0$ .

**Corollary.** *Let  $D$  be a bounded domain. If  $\mathbb{C}^* \setminus D$  consists of finitely many continua, then every point of  $\partial D$  is a regular boundary point.*

Now we can prove the Riemann mapping theorem. Suppose  $D$  is a simply connected domain and that  $\mathbb{C}^* \setminus D$  has at least two points. As in Section XI.3, we can use the logarithm function (or the square root function) and a fractional linear transformation to map  $D$  conformally onto a bounded domain. Indeed, let  $\zeta_0, \zeta_1 \in \partial D$ , and let  $f(z)$  be an analytic branch of  $\log((z - \zeta_0)/(z - \zeta_1))$  in  $D$ . If  $w_0 = f(z_0)$  for some  $z_0 \in D$ , then the image  $f(D)$  covers a disk  $D_0$  centered at  $w_0$ . Consequently,  $f(z)$  cannot assume any of the values in  $D_0 + 2\pi i$  on  $D$ , so that  $1/(f(z) - w_0 - 2\pi i)$  maps  $D$  conformally onto a bounded domain.

So we assume that  $D$  is bounded, and we also assume for convenience that  $0 \in D$ . We solve the Dirichlet problem for the function  $\log|\zeta|$  on  $\partial D$ ,

and we obtain a harmonic function  $u(z)$  on  $D$  such that  $u(z)$  tends to  $\log |\zeta|$  as  $z \in D$  tends to any point  $\zeta \in \partial D$ . Since  $D$  is simply connected, the function  $u(z)$  has a harmonic conjugate  $v(z)$  in  $D$ . Define

$$\varphi(z) = ze^{-(u(z)+iv(z))}, \quad z \in D.$$

Then  $\varphi(z)$  is analytic on  $D$ , and  $|\varphi(z)| \rightarrow 1$  as  $z \rightarrow \partial D$ . By the maximum principle,  $|\varphi(z)| < 1$  for  $z \in D$ . Further,  $\varphi(z)$  has only one zero in  $D$ , a simple zero at  $z = 0$ . Fix  $w$  such that  $|w| < 1$ . We apply the argument principle to the domain  $\{|\varphi(z)| < 1 - \varepsilon\}$ , where  $0 < \varepsilon < |w|$ , and we see that  $\varphi(z) - w$  has exactly one zero on  $D$ . Hence  $\varphi(z)$  maps  $D$  conformally onto the open unit disk.

If we analyze this proof and compare it to the proof given in Chapter XI, we see that both proofs depend on the idea of compactness of families of functions, but in other respects they are quite different. In this proof, Harnack's theorem on equicontinuity of uniformly bounded families of harmonic functions replaces the thesis grade version of Montel's theorem. These two compactness theorems are essentially equivalent. The use of regular boundary points and criteria for regularity replace the use of the hyperbolically expanding property of the square root function, and these techniques are quite different.

### Exercises for XV.5

1. Let  $D$  be a bounded domain such that  $\partial D$  consists of two disjoint continua  $E_0$  and  $E_1$ . Let  $u(z)$  be the harmonic function on  $D$  with boundary values  $u(z) = 0$  on  $E_0$  and  $u(z) = 1$  on  $E_1$ , and let  $v(z)$  be a (locally defined) conjugate harmonic function of  $u(z)$ . Show that there is a constant  $\alpha > 0$  such that  $\varphi(z) = e^{\alpha(u(z)+iv(z))}$  is a single-valued analytic function on  $D$ , and  $w = \varphi(z)$  maps  $D$  conformally onto the annulus  $\{1 < |w| < e^\alpha\}$ .
2. Suppose  $D$  is a doubly connected domain in the extended complex plane  $\mathbb{C}^*$ , that is,  $\mathbb{C}^* \setminus D$  consists of two disjoint connected sets. Show that  $D$  can be mapped conformally onto exactly one of (a) the punctured complex plane  $\mathbb{C} \setminus \{0\}$ , (b) the punctured unit disk  $\mathbb{D} \setminus \{0\}$ , or (c) an annulus  $\{1 < |z| < R\}$ . You may use the result of Exercise 1.

## 6. Green's Function for Domains with Analytic Boundary

We say that the domain  $D$  has **analytic boundary** if the boundary of  $D$  consists of a finite number of disjoint simple closed analytic boundary curves. We will focus in this section on bounded domains with analytic boundary. In this case, every point of  $\partial D$  is a regular boundary point, and we can solve the Dirichlet problem for any continuous boundary function. Further, any harmonic function on  $D$  that is constant on a boundary curve

extends harmonically across the boundary curve, by the Schwarz reflection principle.

So let  $D$  be a bounded domain with analytic boundary. Fix  $\zeta \in D$ , and let  $h(z) = \log|z - \zeta|$  for  $z \in \partial D$ . Then  $h(z)$  is a continuous function on  $\partial D$ , and the Perron solution  $\tilde{h}(z)$  is a harmonic function on  $D$  that assumes the boundary values  $\log|z - \zeta|$  continuously on  $\partial D$ . We define **Green's function** for  $D$  with pole at  $\zeta$  to be the function  $g(z)$  on  $D \setminus \{\zeta\}$  given by

$$g(z) = \tilde{h}(z) + \log \left( \frac{1}{|z - \zeta|} \right).$$

It satisfies

$$(6.1) \quad g(z) \text{ is harmonic on } D \setminus \{\zeta\},$$

$$(6.2) \quad g(z) - \log \frac{1}{|z - \zeta|} \text{ is harmonic at } \zeta,$$

$$(6.3) \quad g(z) \rightarrow 0 \text{ as } z \text{ tends to the boundary of } D.$$

Occasionally, it will be convenient to denote Green's function by  $g(z, \zeta)$ , to emphasize the dependence on  $\zeta$ . If  $h(z)$  is another function with the properties (6.1), (6.2), and (6.3), then  $h(z) - g(z)$  is a harmonic function on  $D$  that tends to zero on the boundary of  $D$ . By the maximum principle,  $h(z) - g(z) = 0$  on  $D$ , and  $h(z) = g(z)$ . Thus Green's function with pole at  $\zeta$  is uniquely determined by the properties (6.1), (6.2), and (6.3).

**Example.** Green's function for the unit disk  $\mathbb{D}$  with pole at  $z = 0$  is  $g(z) = \log(1/|z|)$ .

**Example.** Suppose  $D$  is a simply connected domain,  $D \neq \mathbb{C}$ , and let  $\zeta \in D$ . Let  $\varphi(z)$  be the Riemann map of  $D$  onto the open unit disk  $\mathbb{D}$  satisfying  $\varphi(\zeta) = 0$ , and set  $g(z) = -\log|\varphi(z)|$ . Since  $|\varphi(z)| \rightarrow 1$  as  $z \rightarrow \partial D$ ,  $g(z) \rightarrow 0$  as  $z \rightarrow \partial D$ . Since  $\varphi(z)/(z - \zeta)$  is analytic and nonzero at  $z = \zeta$ , we see by taking logarithms that  $g(z) + \log|z - \zeta|$  is harmonic at  $z = \zeta$ , and so  $g(z)$  has a logarithmic pole at  $z = \zeta$ . Thus  $g(z)$  satisfies (6.1), (6.2), and (6.3) above, and  $g(z)$  is Green's function for  $D$  with pole at  $\zeta$ .

**Lemma.** Let  $D$  be a bounded domain with analytic boundary, and let  $g(z)$  be Green's function for  $D$  with pole at  $\zeta \in D$ . Then  $g(z) > 0$  on  $D$ ,  $g(z) = 0$  on  $\partial D$ , and  $g(z)$  reflects harmonically across  $\partial D$  so that  $g(z) < 0$  on the side of  $\partial D$  outside of  $D$ . Further, the directional derivative of  $g(z)$  in the direction of the exterior normal satisfies  $\partial g / \partial n < 0$  on  $\partial D$ .

The positivity follows from the maximum principle. Since  $g(z) \rightarrow +\infty$  as  $z \rightarrow \zeta$ , we can choose  $\varepsilon > 0$  so small that  $g(z) > 0$  for  $|z - \zeta| \leq \varepsilon$ . The maximum principle, applied to the domain  $D \setminus \{|z - \zeta| \leq \varepsilon\}$ , shows that  $g(z) \geq 0$  for all  $z \in D$ . The strict maximum principle guarantees that  $g(z)$  does not assume a minimum value 0 at any point of  $D$ , so that

$g > 0$  on  $D$ . By the Schwarz reflection principle (Section X.2),  $g$  reflects harmonically across the analytic boundary curves in  $\partial D$  and  $g < 0$  on the side of  $\partial D$  outside  $D$ . Thus  $\partial g / \partial n \leq 0$  on  $\partial D$ . Let  $z_0 \in \partial D$ , and let  $f(z)$  be an analytic function defined on a disk centered at  $z_0$  whose real part is  $g(z)$ . Then  $|f'(z_0)| = |\nabla g(z_0)|$ . Since the set  $\{\operatorname{Re} f(z) = \operatorname{Re} f(z_0)\}$  consists of only one analytic arc passing through  $z_0$ ,  $f'(z_0) \neq 0$ , and consequently,  $(\nabla g)(z_0) \neq 0$ . Since  $g = 0$  on  $\partial D$ ,  $(\nabla g)(z_0)$  is a nonzero multiple of the normal vector at  $z_0$ . Thus  $\partial g / \partial n \neq 0$ , and we conclude that there is strict inequality  $\partial g / \partial n < 0$  on  $\partial D$ . The lemma is proved.

Again let  $D$  be a bounded domain with analytic boundary, and let  $g(z)$  be Green's function for  $D$  with pole at  $\zeta \in D$ . Let  $u(z)$  be any harmonic function on  $D$  that extends continuously to  $\partial D$ . Green's third formula (1.3), applied to  $u(z)$  and to Green's function  $g(z)$  with pole at  $\zeta$ , becomes

$$(6.4) \quad u(\zeta) = -\frac{1}{2\pi} \int_{\partial D} u \frac{\partial g}{\partial n} ds.$$

We have established Green's formulae only for functions that are smooth across the boundary, but in the case of an analytic boundary it is easy to argue (Exercise 8) that the formula holds whenever  $u$  is only continuous up to the boundary.

We define **harmonic measure** on  $\partial D$  for  $\zeta \in D$  to be the differential

$$(6.5) \quad d\mu_\zeta(z) = -\frac{1}{2\pi} \frac{\partial g}{\partial n} ds.$$

Since  $\partial g / \partial n < 0$  on  $\partial D$ , harmonic measure is positive, in the sense that  $\int_{\partial D} v(z) d\mu_\zeta(z) \geq 0$  whenever  $v \geq 0$  on  $\partial D$ . If we apply (6.4) to the function  $u(z) = 1$ , we obtain

$$\int_{\partial D} d\mu_\zeta(z) = 1.$$

We say that  $d\mu_\zeta$  is a probability measure on  $\partial D$  in the sense that it is positive and has unit total mass. In terms of harmonic measure, (6.4) reads as follows.

**Theorem.** *Let  $D$  be a bounded domain with analytic boundary, and let  $g(z)$  be Green's function with pole at  $\zeta \in D$ . Define harmonic measure  $d\mu_\zeta(z)$  by (6.5). Then*

$$u(\zeta) = \int_{\partial D} u(z) d\mu_\zeta(z), \quad \zeta \in D,$$

*for every harmonic function  $u(z)$  on  $D$  that extends continuously to  $\partial D$ .*

**Example.** In the case of the unit disk, Green's function with pole at 0 is  $\log(1/|z|)$ . and

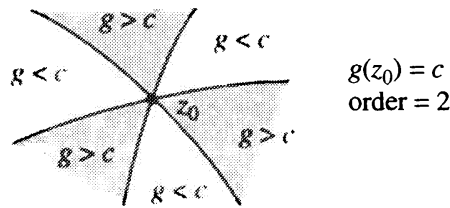
$$\frac{\partial}{\partial n} g(z) = \frac{\partial}{\partial r} \log \frac{1}{r} \Big|_{r=1} = -1, \quad ds = d\theta.$$

Hence harmonic measure on  $\partial\mathbb{D}$  for  $0 \in \mathbb{D}$  is the normalized arc length measure

$$d\mu_0 = \frac{d\theta}{2\pi}.$$

It is straightforward to check (Exercise 2) that the harmonic measure for an arbitrary point  $\zeta \in \mathbb{D}$  is given by the Poisson kernel.

We say that  $z_0$  is a **critical point of Green's function** if  $(\nabla g)(z_0) = 0$ . The value  $g(z_0)$  is called a **critical value** of  $g(z)$ . If  $g(z)$  is the real part of the analytic function  $f(z)$  near  $z_0$ , then  $|f'(z_0)| = |\nabla g(z_0)|$ , so that  $z_0$  is a critical point of  $g(z)$  if and only if  $z_0$  is a zero of  $f'(z)$ . We define the **order of the critical point**  $z_0$  of  $g(z)$  to be the order of zero of  $f'(z)$  at  $z = z_0$ . According to our earlier definition, this is the order of the critical point  $z_0$  of  $f(z)$ . (See Section VIII.5.) Further, if  $z_0$  is a critical point of  $g(z)$  of order  $m$ , then the set  $\{g(z) > g(z_0)\}$  consists of  $m+1$  sector-shaped domains with vertex  $z_0$ , each of aperture  $\pi/(m+1)$ . This makes it clear that no point of  $\partial D$  can be a critical point of  $g(z)$ , because near each  $z_0 \in \partial D$  the set  $\{g(z) > 0\}$  consists of just a half-disk, which can be regarded as a sector of aperture  $\pi$ , so that  $m = 0$ .



**Theorem.** Let  $D$  be a bounded domain with analytic boundary consisting of  $N$  closed curves. Let  $g(z)$  be Green's function for  $D$  with pole at  $\zeta \in D$ . Then  $g(z)$  has  $N - 1$  critical points in  $D$ , counting multiplicities.

The critical points of  $g(z)$  are isolated, since they are the zeros of an analytic function. There are only finitely many critical points of  $g(z)$  in  $D$ , and there are no critical points of  $g(z)$  on  $\partial D$ .

For  $0 \leq r < \infty$ , let  $D_r$  be the set of  $z \in D$  such that  $g(z) > r$  and let  $\Gamma_r = \partial D_r$  be its boundary. For  $r$  large,  $\Gamma_r$  is a circular curve surrounding  $\zeta$ , and  $D_r$  is approximately a disk centered at  $\zeta$ . As  $r$  decreases, from  $+\infty$  to 0, the domains  $D_r$  increase until at  $r = 0$  we reach  $D_0 = D$  and  $\Gamma_0 = \partial D$ . Each  $D_r$  is a domain. Indeed, Green's function is constant on the boundary of each component of  $D_r$ , and Green's function cannot be constant on the

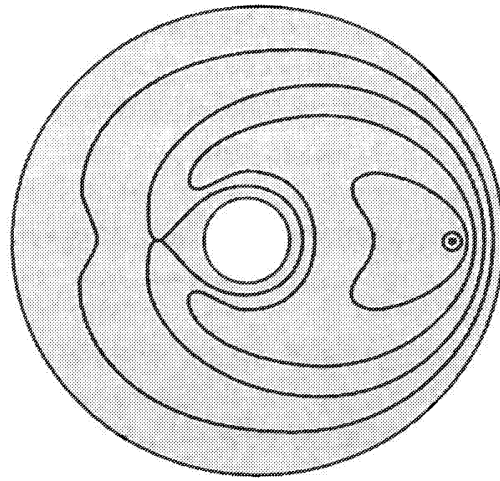
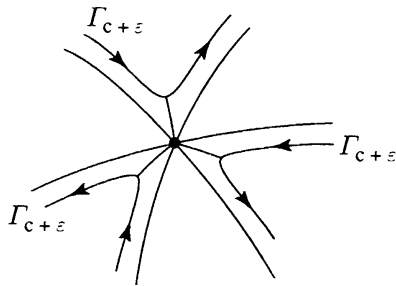
boundary of any component of  $D_r$  on which  $g(z)$  is harmonic, so there can be only one component of  $D_r$ , the component containing the pole  $\zeta$ .

Except at a critical point, each curve in  $\Gamma_r$  is locally the level set of the real part of an analytic function with nonvanishing derivative, hence an analytic curve. Since the curves in  $\Gamma_r$  cannot approach the boundary of  $D$ , each curve in  $\Gamma_r$  is either a simple closed analytic curve, or it is an analytic curve that begins and terminates at (possibly different) critical points. Thus if  $r$  is not a critical value of  $g(z)$ , then  $\Gamma_r$  consists of a finite number of simple closed analytic curves.

Denote by  $U_r$  the complement of  $D_r \cup \Gamma_r$  in the extended complex plane. Each component of  $U_r$  is a simply connected domain bounded by a curve in  $\Gamma_r$ , and there are as many components of  $U_r$  as there are analytic curves in  $\Gamma_r$ . We will count the number of components of  $U_r$  and see how this number changes as  $r$  decreases from  $+\infty$  to 0. The analytic curves in  $\Gamma_r$  move continuously with  $r$  when  $r$  is not a critical value of  $g(z)$ , so the number of components of  $U_r$  remains constant as  $r$  decreases between critical values. We focus on what happens to a component of  $U_r$  when  $r$  crosses over a critical value.

Let  $z_0$  be a critical point of  $g(z)$  of order  $m$ , with critical value  $g(z_0) = c$ , and let  $\varepsilon > 0$  be small. Let  $V$  be the component of  $U_{c+\varepsilon}$  containing  $z_0$ , so that  $\partial V$  is one of the curves in  $\Gamma_{c+\varepsilon}$ . Then  $\{g(z) > c\}$  consists of  $m + 1$  sector-like domains near  $z_0$  with vertices at  $z_0$ . The curve  $\partial V$  approaches  $z_0$  through each of these sectors and then retreats, as in the figure. In each sector we connect  $z_0$  to the nearest point of  $\partial V$  by a straight line segment. We can think of  $V$  as a disk, via the Riemann mapping theorem, and then it is apparent that the  $m + 1$  segments divide  $V$  into  $m + 1$  components, as in the figure. Each of these  $m + 1$  components contains one of the  $m + 1$  sector-like domains with vertex at  $z_0$  where  $g(z) < c$ . We conclude that there are exactly  $m + 1$  components of  $\{g(z) < c\}$  that have  $z_0$  in their boundary. Thus as  $r$  decreases from  $c + \varepsilon$  to  $c - \varepsilon$ , the critical point  $z_0$  of order  $m$  accounts for an increase in the number of components of  $U_r$  by  $m$ . The same is true for all critical points, even though some might correspond to the same critical value. Since  $U_r$  has one component when  $r$  is large, it eventually consists of  $1 + L$  components when  $r$  is small, where  $L$  is the total number of critical points counting multiplicity. Thus  $1 + L = N$ , and the theorem is proved.

The idea of using the sublevel sets of a function such as  $g(z)$  to describe the topology of  $D$  is the starting point for "Morse theory." One way to visualize the procedure is to place the graph of  $g(z)$  in a water reservoir and gradually to drain the water. The curves  $\Gamma_r$  become the shoreline and  $D_r$  an island. As the level of water recedes, the islands  $D_r$  evolve to  $D$  and the shoreline to  $\partial D$ . Dramatic changes occur only when  $r$  passes through a critical value. The figure on the right depicts the level sets of Green's function of an annulus. Note that there is one critical point of order one, which corresponds to a saddle point of the graph of  $g(z)$ .



### Exercises for XV.6

1. Show that for the unit disk  $\mathbb{D}$ , Green's function is given by

$$g(z, \zeta) = -\log \left| \frac{z - \zeta}{1 - \bar{\zeta}z} \right|, \quad z, \zeta \in \mathbb{D}.$$

2. Show by computing the normal derivative of Green's function that harmonic measure for the unit disk  $\mathbb{D}$  is given by the Poisson kernel,

$$d\mu_{\zeta}(e^{i\theta}) = \frac{1 - \rho^2}{1 + \rho^2 - 2\rho \cos(\theta - \varphi)} \frac{d\theta}{2\pi}, \quad \zeta = \rho e^{i\varphi} \in \mathbb{D}.$$

3. Suppose that  $g(z)$  is Green's function for  $D$  with pole at  $\zeta$ , and suppose that  $\varphi(z)$  is a conformal map of a domain  $W$  onto  $D$  such that  $\varphi(\xi) = \zeta$ . Show that  $h(w) = g(\varphi(w))$  is Green's function for  $W$  with pole at  $\xi$ .
4. Suppose that  $D$  is a simply connected domain, mapped by  $\varphi(z)$  conformally onto the open unit disk. Show that Green's function for  $D$  is given by

$$g(z, \zeta) = -\log \left| \frac{\varphi(z) - \varphi(\zeta)}{1 - \overline{\varphi(\zeta)}\varphi(z)} \right|, \quad z, \zeta \in D.$$

5. Let  $D$  be a bounded domain with analytic boundary, and let  $g(z, \zeta)$  be Green's function for  $D$  with pole at  $\zeta$ . Show that if  $u(z)$  is a smooth function on  $D \cup \partial D$ , then

$$u(\zeta) = \int_{\partial D} u(z) d\mu_{\zeta}(z) - \frac{1}{2\pi} \iint_D g(z, \zeta) (\Delta u)(z) dx dy, \quad \zeta \in D$$

6. Let  $D$  be a bounded domain with analytic boundary, let  $g(z)$  be Green's function for  $D$  with pole at  $\zeta$ , and let  $v(z)$  be the (locally defined) harmonic conjugate function of  $g(z)$ . Fix  $r \geq 0$ . Show that



if we orient the curves of the level set  $\{g(z) = r\}$  as boundary curves of the domain  $\{g(z) > r\}$ , then  $v(z)$  is strictly decreasing along each curve in the level set. Show further that the total increase of  $v(z)$  along the curves in the level set  $\{g(z) = r\}$  is  $-2\pi$ .

7. Let  $D$  be a bounded domain with analytic boundary, let  $g(z)$  be Green's function for  $D$  with pole at  $\zeta$ , and let  $v(z)$  be the (locally defined) harmonic conjugate function of  $g(z)$ . Fix one of the boundary curves  $\gamma$  of  $\partial D$ . Show that there is  $\alpha > 0$  such that  $w = e^{\alpha(g+iv)}$  is single-valued in a neighborhood of  $\gamma$  and maps an open set on the side of  $\gamma$  in  $D$  conformally onto an annulus  $\{1 < |w| < 1 + \delta\}$ .
8. Establish formula (6.4) in the case that  $u(z)$  is only continuous on  $\partial D$  by applying the smooth case to the domain  $\{g(z) > \varepsilon\}$  and letting  $\varepsilon \rightarrow 0$ .
9. A point  $(x_0, y_0)$  is a **critical point** of order  $m$  for a smooth function  $u(x, y)$  if all partial derivatives of  $u(x, y)$  with respect to  $x$  and  $y$  of order  $k \leq m$  vanish at  $(x_0, y_0)$ , while at least one partial derivative of order  $m + 1$  does not vanish there. In the case that  $u(x, y)$  is the real part of the analytic function  $f(z)$ , show that  $(x_0, y_0)$  is a critical point of  $u(x, y)$  of order  $m$  if and only if  $z_0 = x_0 + iy_0$  is a critical point of  $f(z)$  of order  $m$ .

## 7. Green's Function for General Domains

Green's function plays a useful role for domains without smooth boundaries and for unbounded domains. In this section we discuss briefly Green's function for general domains.

Let  $D$  be an arbitrary domain, and fix  $\zeta \in D$ . Let  $\{D_n\}$  be an increasing sequence of bounded domains with analytic boundaries such that  $\zeta \in D_1$  and  $\cup D_n = D$ . Let  $g_n(z) = g_n(z, \zeta)$  be Green's function for  $D_n$  with pole at  $\zeta$ . Since  $D_n \subset D_{n+1}$  and  $g_n(z) \leq g_{n+1}(z)$  on  $\partial D_n$ , the maximum principle shows that  $g_n(z) \leq g_{n+1}(z)$  on  $D_n$ . Thus  $\{g_n(z)\}$  is an increasing sequence, and  $g_n(z) + \log |z - \zeta|$  is harmonic on  $D_n$ . According to Harnack's theorem, there are two cases that can occur. Either  $g_n(z)$  tends to  $+\infty$  uniformly on compact subsets of  $D$ , in which case we say that "Green's function for  $D$  does not exist." Or  $g_n(z)$  converges uniformly on compact subsets of  $D \setminus \{\zeta\}$ , in which case we define the limit  $g(z) = g(z, \zeta)$  to be **Green's function** of  $D$  with pole at  $\zeta$ .

**Lemma.** *Green's function  $g(z, \zeta)$  with pole at  $\zeta$  satisfies the following:*

$$(7.1) \quad g(z, \zeta) \text{ is a positive harmonic function on } D \setminus \{\zeta\};$$

$$(7.2) \quad g(z, \zeta) - \log \frac{1}{|z - \zeta|} \text{ is harmonic at } \zeta,$$

(7.3) if  $h(z)$  is a positive harmonic function on  $D \setminus \{\zeta\}$  that has a logarithmic pole at  $\zeta$ , then  $h(z) \geq g(z, \zeta)$ .

Properties (7.1), (7.2), and (7.3) characterize Green's function uniquely.

Properties (7.1) and (7.2) follow from the corresponding properties of  $g_n(z)$ , since  $g_n(z) + \log |z - \zeta|$  converges to  $g(z) + \log |z - \zeta|$  uniformly on a neighborhood of  $\zeta$ . Property (7.3) follows from the maximum principle. Indeed,  $h(z) + \log |z - \zeta|$  dominates  $g_n(z) + \log |z - \zeta|$  on  $\partial D_n$  hence on  $D_n$ , and in the limit we obtain  $h(z) \geq g(z)$  on  $D$ . If  $g_0(z)$  is any other function with the properties (7.1), (7.2), and (7.3), then (7.3) applies to each of  $g(z)$  and  $g_0(z)$ , so  $g(z) \geq g_0(z) \geq g(z)$ , and equality holds.

The uniqueness assertion shows that the definition of  $g(z)$  is independent of the sequence of domains  $\{D_n\}$  increasing to  $D$ . Further, this definition of  $g(z)$  coincides with the definition given in the preceding section where  $D$  is a bounded domain with analytic boundary. In this case we can take  $D_n = D$  for all  $n$ .

**Example.** Let  $D = \mathbb{C}$  be the complex plane, and let  $\zeta = 0$ . We take  $D_n = \{|z| < n\}$ , which has Green's function  $g_n(z) = \log n - \log |z|$ . Since  $g_n(z)$  tends to  $+\infty$  as  $n \rightarrow \infty$ , Green's function does not exist for  $\mathbb{C}$ .

**Lemma.** If  $D$  is a bounded domain, and  $\zeta \in D$ , then Green's function  $g(z, \zeta)$  for  $D$  with pole at  $\zeta$  exists. Further,

$$g(z, \zeta) = \tilde{h}(z) + \log \frac{1}{|z - \zeta|}, \quad z \in D,$$

where  $\tilde{h}(z)$  is the Perron solution of the Dirichlet problem with boundary function  $h(z) = \log |z - \zeta|$ ,  $z \in \partial D$ .

This follows from two applications of the maximum principle. Let  $\{D_n\}$  and  $g_n(z, \zeta)$  be as above. Define  $u_n$  by  $u_n(z) = g_n(z, \zeta)$  for  $z \in D_n$ , and  $u_n(z) = 0$  for  $z \in D \setminus D_n$ . Then  $u_n(z) + \log |z - \zeta|$  is subharmonic on  $D$ , and it coincides with  $\log |z - \zeta|$  on  $\partial D$ . Hence  $u_n(z) + \log |z - \zeta| \in \mathcal{F}_h$ , and  $u_n(z) + \log |z - \zeta| \leq \tilde{h}(z)$ . Passing to the limit on  $D$ , we obtain  $g(z, \zeta) + \log |z - \zeta| \leq \tilde{h}(z)$ .

For the reverse inequality, let  $\varepsilon > 0$ , and let  $u \in \mathcal{F}_h$ . Then  $u(z) - \varepsilon \leq \log |z - \zeta|$  for  $n$  large. From the maximum principle for the harmonic function  $g_n(z, \zeta) + \log |z - \zeta|$  on  $D_n$ , we have  $u(z) - \varepsilon \leq g_n(z, \zeta) + \log |z - \zeta|$  on  $D_n$ . Passing to the limit, first as  $n \rightarrow \infty$  then as  $\varepsilon \rightarrow 0$ , we obtain  $u(z) \leq g(z, \zeta) + \log |z - \zeta|$  on  $D$ . Taking the supremum over such  $u$ , we obtain  $\tilde{h}(z) \leq g(z, \zeta) + \log |z - \zeta|$  on  $D$ . Consequently, equality holds, and the lemma is established.

**Example.** Let  $D = \mathbb{D} \setminus \{0\}$  be the punctured unit disk, and let  $\zeta \in D$ . We have seen that for any continuous function  $h(z)$  on  $\partial D$ , the Perron solu-

tion  $\tilde{h}(z)$  to the Dirichlet problem is the Poisson integral of  $h(e^{i\theta})$ , which coincides with the solution to the Dirichlet problem on  $\mathbb{D}$  with boundary function  $h|_{\partial\mathbb{D}}$ . The preceding theorem then shows that Green's function for  $\mathbb{D}\setminus\{0\}$  coincides with Green's function for  $\mathbb{D}$ . The isolated point  $\{0\}$  of  $\partial D$  is a removable singularity for Green's function.

Now we consider the dependence of  $g(z, \zeta)$  on the parameter  $\zeta$ . We show that the existence of Green's function for a domain  $D$  does not depend on the pole  $\zeta$ . One way to see this is as follows. Let  $\zeta_0 \in D$ , and let  $D_n$  increase to  $D$  as before, with  $\zeta_0 \in D_1$ . Choose  $r > 0$  such that  $\{|z - \zeta_0| \leq r\} \subset D_1$ . Then

$$u_n(z, \zeta) = g_n(z, \zeta) + \log|z - \zeta| - g_n(z, \zeta_0) - \log|z - \zeta_0|$$

is harmonic on  $D_n$ . Further, since Green's function is zero on  $\partial D_n$ ,  $u_n(z, \zeta)$  is bounded in modulus by a constant  $C$  on  $\partial D_n$ , independent of  $n$ , provided that  $|\zeta - \zeta_0| < r/2$ . By the maximum principle,

$$-C \leq g_n(z, \zeta) + \log|z - \zeta| - g_n(z, \zeta_0) - \log|z - \zeta_0| \leq C$$

for all  $z \in D$  and for  $|\zeta - \zeta_0| < r/2$ . This estimate shows that if  $g_n(z, \zeta_0) \rightarrow +\infty$  as  $n \rightarrow \infty$ , then  $g_n(z, \zeta) \rightarrow +\infty$  as  $n \rightarrow \infty$  for  $|\zeta - \zeta_0| < r/2$ , while if  $g_n(z, \zeta_0)$  is bounded, then  $g_n(z, \zeta)$  is bounded for  $|\zeta - \zeta_0| < r/2$ . Thus the sets of  $\zeta$  for which  $g_n(z, \zeta) \rightarrow +\infty$  and for which  $g_n(z, \zeta)$  is bounded form a decomposition of  $D$  into two disjoint open subsets. Since  $D$  is connected, one of these subsets is empty and the other is all of  $D$ . Thus the existence of Green's function does not depend on the pole  $\zeta$ .

In Section XVI.3 we will treat Green's function in the setting of Riemann surfaces, and we will show that the existence of Green's function does not depend on the pole. In Section XVI.4 we will show that Green's function is symmetric, that is, it satisfies  $g(z, \zeta) = g(\zeta, z)$ . In particular,  $g(z, \zeta)$  depends harmonically on the parameter  $\zeta$ .

### Exercises for XV.7

1. Suppose that  $\zeta \in D_0 \subset D$ , and suppose that Green's function  $g(z)$  for  $D$  with pole at  $\zeta$  exists. Show that Green's function  $g_0(z)$  for  $D_0$  with pole at  $\zeta$  exists and satisfies  $g_0(z) \leq g(z)$ .
2. Let  $D$  be a domain for which Green's function  $g(z)$  with pole at  $\zeta$  exists. Show that for each  $r > 0$ , the set of  $z \in D$  satisfying  $g(z) > r$  is a domain, which has Green's function  $g(z) - r$  with pole at  $\zeta$ .
3. Let  $D$  be a domain for which Green's function  $g(z)$  with pole at  $\zeta$  exists. Suppose that  $\varphi(z)$  is a conformal map of a domain  $W$  onto  $D$  such that  $\varphi(\xi) = \zeta$ . Show that  $h(w) = g(\varphi(w))$  is Green's function for  $W$  with pole at  $\xi$ .

4. Show that Green's function for the upper half-plane  $\mathbb{H}$  is

$$g(z, \zeta) = \log \left| \frac{z - \bar{\zeta}}{z - \zeta} \right|, \quad z, \zeta \in \mathbb{H}.$$

5. Show that Green's function for the horizontal strip  $\{|\operatorname{Im} z| < \pi/2\}$  is

$$g(z, \zeta) = \log \left| \frac{e^z + e^{\bar{\zeta}}}{e^z - e^{\zeta}} \right|, \quad -\frac{\pi}{2} < \operatorname{Im} z, \operatorname{Im} \zeta < \frac{\pi}{2}.$$

6. Let  $g(z, \zeta)$  be Green's function for the strip  $\{|\operatorname{Im} z| < \pi/2\}$ . (See the preceding exercise.) Show that for  $\zeta$  fixed and  $\beta > 0$ , the series

$$\sum_{n=-\infty}^{\infty} g(z + \beta n, \zeta)$$

converges uniformly on compact subsets of the strip to a function  $h(z)$  harmonic on the strip except for logarithmic poles at  $\zeta + \beta n$ ,  $-\infty < n < \infty$ . Show that  $h(z)$  is periodic with period  $\beta$ . Show that  $h(\pi i(\log w)/2\alpha)$  is Green's function for the annulus  $\{e^{-\alpha} < |w| < e^{\alpha}\}$  with pole at  $e^{-2\alpha i\zeta/\pi}$ , for  $\alpha = \pi^2/\beta$ .

7. Show that Green's function of  $D$  with pole at  $\zeta$  is the upper envelope of the family of functions  $u(z)$  on  $D$  such that  $u(z) + \log |z - \zeta|$  is subharmonic on  $D$ , and  $u(z) = 0$  off a compact subset of  $D$ .
8. Suppose that Green's function  $g(z)$  for  $D$  exists and that the level set  $\{g(z) = r\}$  does not contain a critical point of  $g(z)$ . Show that each connected component of the level set is either a simple closed analytic curve in  $D$  or an analytic arc, each end of which terminates at an irregular boundary point of  $D$ .
9. Let  $D$  be a domain, let  $\zeta_0 \in \partial D$ , and suppose there is a subharmonic barrier at  $\zeta_0$ . Show that Green's function  $g(z)$  for  $D$  exists, and  $g(z) \rightarrow 0$  as  $z \rightarrow \zeta_0$ . *Remark.* In particular, if Green's function for  $D$  does not exist, then  $\mathbb{C} \setminus D$  does not contain a continuum, so  $\mathbb{C} \setminus D$  is totally disconnected (see Section XII.4).
10. Show that every domain  $D$  in  $\mathbb{C}$  is the increasing union of bounded domains  $D_m$  with analytic boundaries.
11. Let  $D$  be an exterior domain, that is, a domain containing the point at  $\infty$ . If there is a harmonic function  $g(z)$  on  $D$  such that  $g(z) \rightarrow 0$  as  $z \rightarrow \partial D$ , and  $g(z) - \log |z|$  is bounded as  $z \rightarrow \infty$ , we say that  $g(z)$  is **Green's function for  $D$  with pole at  $\infty$** . (a) Show that Green's function with pole at  $\infty$  is unique (if it exists). (b) Show that  $g(z) > 0$  for  $z \in D$ . (c) Show that  $g(z) - \log |z|$  has a limit

as  $z \rightarrow \infty$ . *Remark.* The limit is called **Robin's constant** of  $D$ .

(d) Show that if  $D$  is simply connected, then Robin's constant is  $-\log |\varphi'(\infty)|$ , where  $\varphi(z)$  is the Riemann map of  $D$  onto the open unit disk satisfying  $\varphi(\infty) = 0$ , and  $\varphi'(\infty) = \lim_{z \rightarrow \infty} z\varphi(z)$ .

12. Show that the extended complex plane  $\mathbb{C}^*$  has no Green's function with pole at  $\infty$ , in the sense that there is no harmonic function on the complex plane such that  $g(z) - \log |z|$  is bounded as  $z \rightarrow \infty$ .

# Riemann Surfaces

Our goal in this chapter is to prove the uniformization theorem for Riemann surfaces and to indicate its usefulness as a tool in complex analysis. We begin in Sections 1 and 2 by defining Riemann surfaces, providing examples, and showing how various local notions as analytic function, meromorphic function, and harmonic function carry over to Riemann surfaces. In Sections 3 and 4 we define Green's function for Riemann surfaces and show that Green's function is symmetric. In Section 5 we show that every Riemann surface has bipolar Green's functions. We prove the uniformization theorem in Section 6. The proof depends on Green's function when it exists and on bipolar Green's function otherwise. In Section 7 we define covering spaces and covering maps, and we state several results that indicate the power of the uniformization theorem.

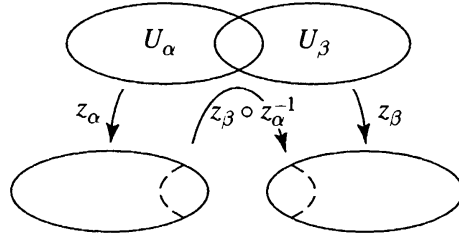
## 1. Abstract Riemann Surfaces

Roughly speaking, a Riemann surface is a space that is locally the same as the complex plane. A near sighted person standing on the surface could not distinguish it from the complex plane. We will use the following definition.

A **Riemann surface** is a set  $R$  with a collection of subsets  $\{U_\alpha\}$  and complex-valued functions  $z_\alpha(p)$ ,  $p \in U_\alpha$ , such that

- (1.1) each  $z_\alpha(p)$  is a one-to-one function from  $U_\alpha$  onto a domain  $z_\alpha(U_\alpha)$  in the complex plane;
- (1.2) each composition  $z_\beta \circ z_\alpha^{-1}$  is analytic wherever it is defined, that is, from  $z_\alpha(U_\alpha \cap U_\beta)$  to  $z_\beta(U_\alpha \cap U_\beta)$ ;
- (1.3)  $R$  is connected, that is, for any two points  $p$  and  $q$  in  $R$ , there is a finite collection of indices  $\alpha_1, \dots, \alpha_m$  such that  $p \in U_{\alpha_1}$ ,  $q \in U_{\alpha_m}$ , and  $U_{\alpha_j} \cap U_{\alpha_{j+1}}$  is nonempty for  $1 \leq j < m$ ;
- (1.4)  $R$  is a "Hausdorff space," that is, if  $p_0$  and  $p_1$  are distinct points of  $R$ , and if  $p_0 \in U_\alpha$  and  $p_1 \in U_\beta$ , then there are (small) disks  $D_0$  and  $D_1$  centered at  $z_\alpha(p_0)$  and  $z_\beta(p_1)$ , respectively, such that  $\pi_\alpha^{-1}(D_0)$  and  $\pi_\beta^{-1}(D_1)$  are disjoint subsets of  $R$ .

We refer to  $U_\alpha$  as a **coordinate patch** and to  $z_\alpha(p)$  as the **coordinate map** on  $U_\alpha$ , mapping  $U_\alpha$  onto  $z_\alpha(U_\alpha)$ . The composition  $z_\beta \circ z_\alpha^{-1}$  corresponds to a change of variable. If  $D_0$  is a disk in  $z_\alpha(U_\alpha)$ , we refer to  $W_0 = z_\alpha^{-1}(D_0)$  as a **coordinate disk** in  $R$ . The coordinate function  $z_\alpha$  maps  $W_0$  one-to-one onto  $D_0$ .



**Example.** Each domain  $D$  in the complex plane can be regarded as a Riemann surface. We require only one coordinate patch  $U_1 = D$ , with coordinate map the identity function  $z_1(\zeta) = \zeta$ .

**Example.** The simplest nontrivial Riemann surface is the extended complex plane  $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$ . This can be coordinatized by two coordinate patches  $U_0 = \mathbb{C}$  and  $U_1 = \mathbb{C}^* \setminus \{0\}$ , with coordinate maps  $z_0(\zeta) = \zeta$  and  $z_1(\zeta) = 1/\zeta$ . In this case,  $z_0(U_0 \cap U_1) = \mathbb{C} \setminus \{0\}$ , and the change of coordinate variable is given by

$$(z_1 \circ z_0^{-1})(\zeta) = 1/\zeta, \quad \zeta \in \mathbb{C} \setminus \{0\},$$

which is analytic.

**Example.** Let  $\omega_1$  and  $\omega_2$  be two complex numbers that do not lie on the same line through the origin, and let  $L$  be the lattice they generate, consisting of all linear combinations  $m\omega_1 + n\omega_2$  where  $m$  and  $n$  are integers. We denote by  $T = \mathbb{C}/L$  the torus determined by  $L$ . Points of  $T$  can be regarded as congruence classes  $\zeta + L$ , where  $\zeta \in \mathbb{C}$ . Choose  $\varepsilon > 0$  so small that all nonzero lattice points satisfy  $|m\omega_1 + n\omega_2| > \varepsilon$ . For each complex number  $\lambda \in \mathbb{C}$ , let  $U_\lambda$  be the subset of  $T$  defined by

$$U_\lambda = \{\zeta + L : |\zeta - \lambda| < \varepsilon\}.$$

The coordinate map  $z_\lambda : U_\lambda \rightarrow \mathbb{C}$  is defined by  $z_\lambda(\zeta + L) = \zeta$  for  $|\zeta - \lambda| < \varepsilon$ . It maps  $U_\lambda$  onto the disk  $\{|\zeta - \lambda| < \varepsilon\}$ . The condition on  $\varepsilon$  guarantees that the coordinate map  $z_\lambda$  is one-to-one. A change of coordinate  $z_\alpha \circ z_\lambda^{-1}$  is the identity, wherever defined, hence analytic.

**Example.** Consider the Riemann surface  $R$  of  $w = \sqrt{z}$  as constructed in Section I.4. It is obtained from two copies  $S_1, S_2$  of the slit  $z$ -plane  $\mathbb{C} \setminus (-\infty, 0]$  by identifying edges appropriately. We can make  $R$  into an abstract Riemann surface with four coordinate patches  $U_1, U_2, U_3, U_4$ , where  $U_1$  is  $S_1$  (without the edges) and  $z_1$  is the natural map of  $S_1$  onto  $\mathbb{C} \setminus (-\infty, 0]$ ; similarly for  $U_2$  and  $z_2$ ;  $U_3$  consists of the top half of  $S_1$ , the

bottom half of  $S_2$ , and the slit along which they are joined, and  $z_3$  is the natural map of  $U_3$  onto  $\mathbb{C} \setminus [0, +\infty)$ ; and similarly for  $U_4$ . In this case, the change of variable corresponding to two coordinate maps is the identity function wherever defined, hence analytic.

**Example.** The Riemann surface  $R$  of  $w = \log z$ , as constructed in Section I.5, can be made into an abstract Riemann surface in the same way. In this case,  $R$  is obtained from an infinite sequence of copies of the slit plane  $\mathbb{C} \setminus (-\infty, 0]$  by identifying edges of slits appropriately. This time we use infinitely many coordinate patches, one for each sheet and one to cover each line where two sheets are attached.

Topological notions and notions of convergence and continuity are carried over to a Riemann surface  $R$  by referring them to the complex plane via the coordinate maps. We define a subset  $W$  of  $R$  to be **open** if for all  $\alpha$  the image  $z_\alpha(W \cap U_\alpha)$  is an open subset of the complex plane. Since a change of variable  $z_\beta \circ z_\alpha^{-1}$  maps open sets to open sets, a subset of a coordinate patch  $U_\alpha$  is open in  $R$  if and only if its image under the single coordinate map  $z_\alpha$  is open.

We say that a sequence  $\{p_n\}$  of points of  $R$  **converges** to  $p \in R$  if for any coordinate patch  $U_\alpha$  containing  $p$ , the  $p_n$ 's eventually belong to  $U_\alpha$ , and  $z_\alpha(p_n)$  converges to  $z_\alpha(p)$ . The "Hausdorff" condition (1.4) asserts simply that the limit of a convergent sequence is unique. It is possible to construct pathological spaces that satisfy the other three axioms (1.1)-(1.3) but for which limits of sequences are not unique.

A subset  $K$  of  $R$  is **compact** if  $K$  is contained in a finite union of coordinate patches, and every sequence in  $K$  has a subsequence that converges to a point in  $K$ . This occurs if and only if  $K$  can be expressed as a finite union  $K = K_1 \cup \cdots \cup K_m$ , where each  $K_j$  is a compact subset of a single coordinate patch.

A complex-valued function  $f(p)$  defined on  $R$  is **continuous at  $q$**  if  $f \circ z_\alpha^{-1}$  is continuous at  $z_\alpha(q)$  for any coordinate patch  $U_\alpha$  containing  $q$ . This occurs if and only if  $f(p_n) \rightarrow f(q)$  whenever  $p_n$  converges to  $q$ . Similarly, a function  $g$  from a subset  $E$  of  $\mathbb{R}^n$  to  $R$  is **continuous** if for any coordinate patch  $U_\alpha$ , the composition  $z_\alpha \circ g$  is continuous on  $g^{-1}(U_\alpha)$ .

A **path** in  $R$  from  $p_0$  to  $p_1$  is a continuous function  $\gamma(t)$ ,  $0 \leq t \leq 1$ , from the unit interval  $[0, 1]$  into  $R$  such that  $\gamma(0) = p_0$  and  $\gamma(1) = p_1$ . The path is **smooth** if the compositions  $z_\alpha \circ \gamma$  are smooth wherever defined. Piecewise smooth paths and analytic curves are defined similarly.

An open subset  $D$  of a Riemann surface is **connected** if any two points of  $D$  can be joined by a path in  $D$ . The connectivity hypothesis (1.3) in the definition of Riemann surface is equivalent to requiring that the surface be connected. Indeed, since each  $z_\alpha(U_\alpha)$  is a domain, any two points of  $U_\alpha$  can be joined by a path, and by concatenating a finite number of paths we can join any two points of  $R$ .



**Example.** A connected open subset of a Riemann surface can be regarded as a Riemann surface, by restricting the coordinate maps to the subset. We refer to the connected open subset as a **subsurface** of the Riemann surface, or as a **domain** on the surface.

In addition to properties involving convergence and continuity, any “local” property that is invariant under an analytic change of variable can be carried over to a Riemann surface by means of the coordinate maps. We list several such properties.

A complex-valued function  $f(p)$  defined in an open subset  $V$  of  $R$  is **analytic** if  $f \circ z_\alpha^{-1}$  is analytic on  $z_\alpha(V \cap U_\alpha)$  for each coordinate patch  $U_\alpha$ . A function  $f : V \rightarrow \mathbb{C}^*$  is **meromorphic** if  $f \circ z_\alpha^{-1}$  is meromorphic on each  $z_\alpha(V \cap U_\alpha)$ . The function  $f(p)$  is **analytic at**  $p_0 \in R$  (respectively **meromorphic at**  $p_0 \in R$ ) if it is analytic (respectively meromorphic) in some coordinate disk containing  $p_0$ .

The order of a zero of an analytic function at a point is invariant under an analytic change of variable. Thus we can talk about the zeros and their orders of an analytic function on a Riemann surface. Similarly, the orders of poles of meromorphic functions on a Riemann surface are well-defined.

**Example.** For a domain in the complex plane, regarded as a Riemann surface, the definitions of analytic and meromorphic functions reduce to the usual definitions.

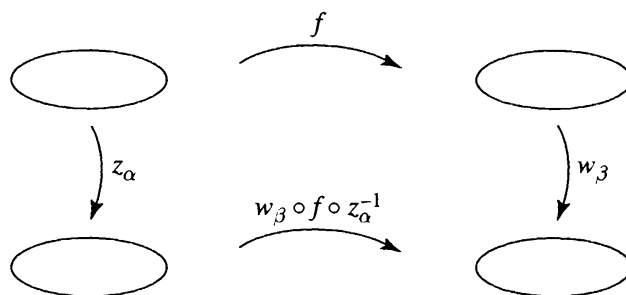
**Example.** Since the coordinate map for  $\mathbb{C}^* \setminus \{0\}$  is  $z_1(\zeta) = 1/\zeta$ , the analytic functions on a domain  $R$  on the Riemann sphere  $\mathbb{C}^*$  are the functions  $f(\zeta)$  that are analytic on  $R \cap \mathbb{C}$ , such that  $g(\zeta) = f(1/\zeta)$  is analytic at  $\zeta = 0$ . Thus the definition of a function being analytic at  $\infty$  when we regard  $R$  as an abstract Riemann surface is the same as the definition given in Section V.5 for analyticity at  $\infty$ . Similarly, the definition of a function being meromorphic at  $\infty$  when we regard  $R$  as an abstract Riemann surface is the same as the definition of being meromorphic at  $\infty$  given in Section VI.3.

**Example.** We saw in Section I.4 that the function  $\sqrt{z}$  can be defined continuously on its Riemann surface  $R$ , by defining it to be the principal branch on one sheet and the negative of the principal branch on the second sheet. Thus defined, the function  $\sqrt{z}$  is actually analytic on  $R$ . On each of the coordinate patches  $U_1, U_2, U_3, U_4$ , the function corresponds to an analytic branch of  $\sqrt{z}$  on a slit plane, with respect to the local coordinate map.

**Example.** Each doubly periodic meromorphic function  $f(z)$  on the complex plane with periods  $\omega_1, \omega_2$  determines a meromorphic function  $g(p)$  on the torus  $T = \mathbb{C}/L$  by  $g(z + L) = f(z)$ . Every meromorphic function on the torus arises in this manner from a doubly periodic function on the complex plane.

We have defined analyticity for complex-valued functions on a Riemann surface. We can also define a notion of analyticity for a function from one Riemann surface to another.

Let  $R$  and  $S$  be two Riemann surfaces. A function  $f : R \rightarrow S$  is **analytic at**  $p \in R$  if for any coordinate patch  $U_\alpha$  in  $R$  containing  $p$  with coordinate map  $z_\alpha$ , and any coordinate patch  $V_\beta$  in  $S$  containing  $f(p)$  with coordinate map  $w_\beta$ , the composition  $w_\beta \circ f \circ z_\alpha^{-1}$  is analytic at  $z_\alpha(p)$ . We refer to  $f$  also as an **analytic map** from  $R$  to  $S$ .



**Example.** The analytic functions from a Riemann surface  $R$  to the Riemann sphere  $\mathbb{C}^*$  are the meromorphic functions on  $R$ , together with the constant function  $\infty$  on  $R$  (Exercise 4).

Technically, our definition of Riemann surface depends not only on the set  $R$  but the specific coordinate patches  $\{U_\alpha\}$  and coordinate maps  $z_\alpha$ . We will regard two Riemann surfaces as being the “same” if the underlying sets are the same and the identity map between them is analytic and has an analytic inverse. This occurs if and only if they have the same analytic functions, and in this case we can use as a coordinate map any one-to-one analytic function from an open subset  $U$  of  $R$  onto a domain in the complex plane. This allows us to focus on coordinate disks when convenient.

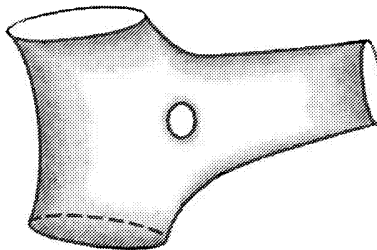
We say that two Riemann surfaces  $R$  and  $S$  are **conformally equivalent** if there is an analytic map  $f : R \rightarrow S$  that is one-to-one and onto. In this case, the inverse map  $f^{-1} : S \rightarrow R$  is also analytic.

**Example.** The Riemann surface of  $\sqrt{z}$  is conformally equivalent to the punctured complex plane  $\mathbb{C} \setminus \{0\}$ . The analytic map giving the equivalence is the continuous extension of the function  $f(z) = \sqrt{z}$  to the Riemann surface.

Let  $S$  be a Riemann surface. A subsurface  $R$  of  $S$  with boundary  $\partial R$  is a **finite bordered subsurface** if  $R \cup \partial R$  is a compact subset of  $S$ , and  $\partial R$  consists of a finite number of disjoint simple closed analytic curves in  $S$ . Thus for each  $p_0 \in \partial R$  there is a coordinate disk  $U$  in  $S$  with coordinate map  $z(p)$  that maps  $U \cap R$  onto a semidisk in the upper half-plane and maps  $U \cap \partial R$  onto an interval on the real axis. The analytic structure on  $R \cup \partial R$  is actually independent of the ambient surface  $S$ , and is determined

by the coordinate disks for points of  $R$  and coordinate semidisks for points of  $\partial R$ . We refer to  $R \cup \partial R$  as a **finite bordered Riemann surface**.

**Example.** Let  $S$  be a compact Riemann surface, and let  $D_1, \dots, D_N$  be  $N$  disjoint closed coordinate disks in  $S$ . If we punch out the closed disks, we obtain a finite bordered Riemann surface  $R = S \setminus \bigcup D_j$  with border  $\partial R$  consisting of the  $N$  analytic boundary curves  $\partial D_j$ ,  $1 \leq j \leq N$ . It turns out (Exercise 2.5) that every finite bordered Riemann surface can be obtained in this way, from a compact surface by excising a finite number of coordinate disks.



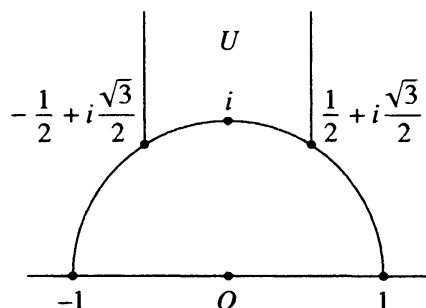
### Exercises for XVI.1

1. Define the Riemann surface  $R$  of  $\log z$  in terms of explicit coordinate patches. Define explicitly the function on  $R$  determined by  $\log z$ . Show that it is a one-to-one analytic map of  $R$  onto the complex plane  $\mathbb{C}$ .
2. What is the smallest number of coordinate patches necessary to make a torus into a Riemann surface?
3. Show (a) that the residue of a meromorphic function  $f(p)$  at a pole  $p_0 \in R$  depends on the choice of the coordinate at  $p_0$ , and (b) that whether the residue of a meromorphic function  $f(p)$  vanishes at a pole  $p_0 \in R$  is independent of the choice of the coordinate at  $p_0$ .
4. Show that the analytic maps from a Riemann surface  $R$  to the Riemann sphere  $\mathbb{C}^*$  are the meromorphic functions on  $R$  and the constant function  $\infty$ .
5. Let  $\omega \neq 0$ , and let  $\mathbb{Z}\omega$  be the integral multiples of  $\omega$ . Let  $R$  be the set of congruence classes  $z + \mathbb{Z}\omega$ ,  $z \in \mathbb{C}$ . Show that  $R$  is a Riemann surface that is conformally equivalent to the punctured plane  $\mathbb{C} \setminus \{0\}$ .
6. Consider the Riemann surface  $R$  of  $\sqrt{z}$  as described in the example in this section. Let  $S = R \cup \{0, \infty\}$ . Define a coordinate disk  $V_0$  containing 0 to be the point 0 together with the set of  $z \in S_1 \cup S_2$  such that  $|z| < 1$ , and define the coordinate map  $\pi_0 : V_0 \rightarrow \Delta$  by  $\pi_0(z) = \sqrt{z}$ ,  $\pi_0(0) = 0$ . Similarly, define a coordinate disk  $V_1$  containing  $\infty$  to be the point  $\infty$  together with the set of  $z \in S_1 \cup S_2$

- such that  $|z| > 1$ , with coordinate map  $\pi_1 : V_1 \rightarrow \mathbb{D}$ ,  $\pi_1(z) = 1/\sqrt{z}$ ,  $\pi_1(\infty) = 0$ . Show that  $S$  is a Riemann surface. Show that the function  $\sqrt{z}$  on  $R$  extends to a one-to-one analytic map of  $S$  onto the Riemann sphere  $\mathbb{C}^*$ .
7. Suppose that  $f : R \rightarrow S$  is a nonconstant analytic map. (a) Show that  $f(R)$  is an open subset of  $S$ . (b) Show that if  $R$  is compact, then  $f(R) = S$ . (c) Use (b) to prove the fundamental theorem of algebra, that every nonconstant polynomial  $P(z)$  has a zero.
  8. Show that if  $U$  and  $V$  are disjoint open subsets of a Riemann surface  $R$  such that  $R = U \cup V$ , then either  $U = R$  or  $V = R$ . Use this to show that the uniqueness principle for meromorphic functions holds on a Riemann surface  $R$ , that is, if  $f(p)$  and  $g(p)$  are two meromorphic functions on  $R$  such that  $f(p) = g(p)$  for  $p$  in some nonempty open subset of  $R$ , then  $f(p) = g(p)$  for all  $p \in R$ .
  9. Let  $f(\zeta)$  be analytic at  $\zeta = \zeta_0$ , and let  $f_{\zeta_0}$  be its power series expansion at  $\zeta_0$ . Let  $R$  be the set of all pairs  $(\zeta, f_\zeta)$ , where  $\zeta \in \mathbb{C}$ ,  $f_\zeta$  is a power series expansion centered at  $\zeta$  with a positive radius of convergence, and  $f_\zeta$  is the analytic continuation of  $f_{\zeta_0}$  along some path from  $\zeta_0$  to  $\zeta$ . Define a coordinate patch containing  $(\zeta, f_\zeta)$  to consist of all pairs  $(\xi, f_\xi)$  such that  $|\xi - \zeta| < \varepsilon$  and  $f_\xi$  is the power series expansion of  $f_\zeta(z)$  about  $z = \xi$ . Define the coordinate map on this coordinate patch by  $\pi(\xi, f_\xi) = \xi$ . (a) Show that  $R$  is a Riemann surface. (b) Show that the projection function  $\pi(\xi, f_\xi) = \xi$  is analytic on  $R$ . (c) Show that  $F(\xi, f_\xi) = f_\xi(\xi)$  is analytic on  $R$  and that in terms of the above coordinate map at  $\zeta_0$  it coincides with  $f(\zeta)$  in a coordinate disk centered at  $\zeta_0$ . (d) Show that if  $P(\zeta, w)$  is a polynomial in  $\zeta$  and  $w$  such that  $P(\zeta, f(\zeta)) = 0$  for  $|\zeta - \zeta_0| < \varepsilon$ , then  $P(\xi, f_\xi(\xi)) = 0$  for all  $(\xi, f_\xi) \in R$ .
  10. Let  $x_1 < x_2 < \cdots < x_n$  be  $n$  consecutive points on the real axis. Let  $\zeta_0$  be distinct from the  $x_j$ 's, and let  $f(\zeta)$  be a branch of  $\sqrt{(\zeta - x_1) \cdots (\zeta - x_n)}$  defined at  $\zeta_0$ . Describe the Riemann surface  $R$  of  $f(\zeta)$  constructed in the preceding exercise, and show that it is the same as the Riemann surface described in Exercise I.7.7. Show that  $\pi(\xi, f_\xi) = \xi$  is a two-to-one map of  $R$  onto  $\mathbb{C} \setminus \{x_1, \dots, x_n\}$ . Show that the surface  $R$  is equivalent to a compact surface  $S$  with a finite number of punctures. *Remark.* The Riemann surface  $S$  is a hyperelliptic Riemann surface.
  11. Let  $R$  be a finite bordered Riemann surface, with border  $\partial R$ . Let  $\tilde{R}$  be a duplicate copy of  $R$ , and denote by  $\tilde{p}$  the point in  $\tilde{R}$  corresponding to  $p \in R$ . Let  $S = R \cup \tilde{R} \cup \partial R$ . Define  $\tau : S \rightarrow S$  by  $\tau(p) = p$  if  $p \in \partial R$  and by  $\tau(p) = \tilde{p}$ ,  $\tau(\tilde{p}) = p$  if  $p \in R$ .

- (a) Show that  $S$  can be made into a compact Riemann surface with  $R$  as a subsurface so that  $\tau$  is anticonformal, that is,  $f(p)$  is analytic on an open set  $U$  if and only if  $\overline{f(\tau(p))}$  is analytic on  $\tau(U)$ . *Remark.* The surface  $S$  is the **doubled surface** of  $R$ , and  $\tau$  is the reflection in  $\partial R$ .
  - (b) Show that the doubled surface of the unit disk is the Riemann sphere  $\mathbb{C}^*$ .
  - (c) What is the doubled surface of an annulus?
  - (d) Show that each meromorphic function  $f$  on  $S$  has the form  $f = g + ih$ , where  $g$  and  $h$  are meromorphic functions on  $S$  that are real-valued on  $\partial R$ .
  - (e) Show that if  $f$  is meromorphic on  $R$ , and if  $\operatorname{Im} f(p) \rightarrow 0$  as  $p \rightarrow \partial R$ , then  $f$  extends to be meromorphic on  $S$ , and the extension is analytic and real-valued on  $\partial R$ .
12. Show that a hyperelliptic Riemann surface (Exercise 10) is the double of a finite bordered subsurface. Specify the anticonformal reflection  $\tau$ .
13. For  $\tau$  in the open upper half-plane  $\mathbb{H}$ , denote by  $L_\tau$  the lattice  $\mathbb{Z} + \mathbb{Z}\tau$  generated by 1 and  $\tau$ , and denote the Riemann surface  $\mathbb{C}/L_\tau$  by  $T_\tau$ .
- (a) Show that the Riemann surface  $T = \mathbb{C}/L$  in the example in this section is conformally equivalent to the Riemann surface  $T_\tau$  for some  $\tau \in \mathbb{H}$ . *Hint.* Take  $\tau = \pm\omega_1/\omega_2$ , where the sign is chosen so that  $\operatorname{Im} \tau > 0$ .
  - (b) Show that  $T_\tau$  is conformally equivalent to  $T_{\tau'}$  if and only if there is a fractional linear transformation of the form  $f(z) = (az + b)/(cz + d)$  where  $a, b, c, d$  are integers satisfying  $ad - bc = 1$ , such that  $f(\tau) = \tau'$ . *Remark.* The matrix with entries  $a, b, c, d$  is called a **unimodular matrix**. The unimodular matrices form a group, which is the special linear group  $\operatorname{SL}(2, \mathbb{R})$ .
14. Let  $\mathcal{U}$  denote the fractional linear transformations  $f(z) = (az + b)/(cz + d)$  corresponding to unimodular matrices. Let  $U$  be the domain in the upper half-plane consisting of points  $z \in \mathbb{H}$  satisfying  $|z| > 1$  and  $-\frac{1}{2} < \operatorname{Re} z < \frac{1}{2}$ .
- (a) Show that  $g(z) = -1/z$  belongs to  $\mathcal{U}$ , and  $g(z)$  has a fixed point at  $i$ . Sketch the image of  $U$  under  $g(z)$ .
  - (b) Show that  $h(z) = 1 - 1/z$  belongs to  $\mathcal{U}$ , and  $h(z)$  has a fixed point at the vertex  $(1 + \sqrt{3}i)/2$  of  $U$ . Sketch the images of  $U$  under  $h(z)$  and  $(h \circ h)(z)$ .
  - (c) Show that the images of  $U$  and its boundary curves under the fractional linear transformations in  $\mathcal{U}$  fill out the upper half-plane. *Hint.* Show that the images fill out an open set containing  $U \cup \partial U$ .

- (d) Show that if  $f \in \mathcal{U}$ , then either  $f(z) = z + m$  for some integer  $m$ , or  $f(z)$  maps the outside of the circle  $|z - d/c| = 1/|c|$  onto the inside of a circle centered on the real axis of the same radius  $1/|c|$ . *Remark.* The circle  $|z - d/c| = 1/|c|$  is the **isometric circle** of  $f(z)$ .
- (e) Show that if  $z, w \in U \cup \partial U$  satisfy  $w = f(z)$  for some  $f \in \mathcal{U}$  other than the identity transformation, then either  $w = z \pm 1$  and both points lie on opposite vertical sides of  $U$ , or  $w = -1/z$  and both points lie on the unit circle. *Remark.* Thus the set  $E$  consisting of  $U$  and the points  $z \in \partial U$  satisfying  $\operatorname{Re} z \leq 0$  is a “fundamental domain” for  $\mathcal{U}$ , in the sense that for each  $w \in \mathbb{H}$  there is a unique  $z \in E$  such that  $w = f(z)$  for some  $f \in \mathcal{U}$ . The images of  $U$  under the functions in  $\mathcal{U}$  form a “tiling” of  $\mathbb{H}$ . *Remark.* With composition as the operation,  $\mathcal{U}$  is a group, called the **modular group**. If  $A$  is a unimodular matrix, then both  $\pm A$  give rise to the same  $f \in \mathcal{U}$ , and every  $f \in \mathcal{U}$  arises from exactly two unimodular matrices. Thus the modular group is isomorphic to the quotient  $\operatorname{SL}(2, \mathbb{R})/\{\pm I\}$ .



## 2. Harmonic Functions on a Riemann Surface

In this section we adapt the Perron procedure to a Riemann surface to solve the Dirichlet problem. The development proceeds very much along the same lines as for planar domains. The success of the procedure depends on the fact that the compactness theorems for harmonic functions are local, as are the notions of subharmonic function and subharmonic barrier.

A real-valued function  $u(p)$  on a Riemann surface  $R$  is **harmonic** at  $p_0 \in R$  if there is a coordinate disk containing  $p_0$  such that  $u(p)$  is a harmonic function of the coordinate  $z = z(p)$  on the coordinate disk. This occurs if and only if  $u(p)$  is the real part of an analytic function on the coordinate disk. Thus the definition of harmonicity is independent of the coordinate map.

The local arguments involving harmonic functions on parameter disks, together with compactness and connectedness arguments, allow us to carry over to Riemann surfaces the compactness theorems for families of har-

monic functions from Section XV.3. One specific result we will use is the following version of Harnack's estimate.

**Lemma.** *For each compact subset  $K$  of a Riemann surface  $R$ , there is a constant  $C > 0$  such that*

$$\frac{1}{C} \leq \frac{u(p)}{u(q)} \leq C, \quad p, q \in K,$$

*for every positive harmonic function on  $R$ .*

If  $p$  belongs to a closed subdisk of a coordinate disk that has  $q$  as its center, this is simply Harnack's inequality. If we replace  $C$  by  $C^2$ , we obtain an estimate of the same form for arbitrary points  $p$  and  $q$  in a fixed closed subdisk of a coordinate disk. For the general case we cover  $K$  by a connected set that is a finite union of closed subdisks of coordinate disks, and we apply the estimate to successive points in a chain  $p = p_0, p_1, \dots, p_n = q$ , where each pair  $p_j, p_{j+1}$  belong to the same disk. (See Exercises XV.3.4-5.)

Now let  $R$  be a Riemann surface, and let  $W$  be an open subset of  $R$ . A continuous function  $u : W \rightarrow [-\infty, +\infty)$  is **subharmonic** on  $W$  if each  $p \in W$  belongs to a coordinate patch on which  $u(p)$  is a subharmonic function of the coordinate  $z = z(p)$ . (We could allow subharmonic functions to be only "upper semicontinuous," but for our goals this leads to no gain.)

**Theorem (Strict Maximum Principle).** *If  $u$  is a subharmonic function on a Riemann surface  $R$ , and  $u$  attains its maximum at some point of  $R$ , then  $u$  is constant on  $R$ .*

The proof of the strict maximum principle for subharmonic functions (Section XV.2) carries over to Riemann surfaces. Let  $E$  be the set of  $p \in R$  where  $u$  attains its maximum. The local argument used in the plane shows that  $E$  is open, while the continuity of  $u$  implies that  $R \setminus E$  is also open. The connectedness of  $R$  then implies that either  $E = R$  or  $E$  is empty. In the case at hand, we must have  $E = R$ , and then  $u$  is constant.

Since continuous functions on a compact set attain their maximum, we obtain as an immediate consequence the following version of the maximum principle.

**Theorem (Maximum Principle).** *Let  $u$  be a subharmonic function on a Riemann surface  $R$ . If  $u(p) \leq c$  for  $p \in R \setminus K$ , where  $K$  is a compact subset of  $R$ , then  $u(p) \leq c$  for all  $p \in R$ .*

Now we may solve the Dirichlet problem on a Riemann surface by the Perron process of taking an upper envelope of subsolutions. To make this explicit, it is convenient to formalize the Perron procedure.

Let  $W$  be an open subset of the Riemann surface  $R$ . We say that a nonempty family  $\mathcal{F}$  of subharmonic functions on  $W$  is a **Perron family of subharmonic functions** if the following two conditions are satisfied:

- (2.1) if  $u, v \in \mathcal{F}$ , then  $\max(u, v) \in \mathcal{F}$ , where  $\max(u, v)$  is the maximum of  $u$  and  $v$ ,
- (2.2) if  $u \in \mathcal{F}$ , and  $D_0$  is a coordinate disk in  $W$  such that  $u$  is finite  $\partial D_0$ , then the function  $v$  defined to be  $u$  on  $W \setminus D_0$  and the harmonic extension of  $u|_{\partial D_0}$  on  $D_0$  is in  $\mathcal{F}$ .

The proof given in Section XV.4 carries over to show the following.

**Theorem.** *Let  $\mathcal{F}$  be a Perron family of subharmonic functions on a domain  $W$  on a Riemann surface  $R$ , and let  $u$  be the upper envelope of the family  $\mathcal{F}$ ,*

$$u(p) = \sup\{v(p) : v \in \mathcal{F}\}.$$

*Then either  $u$  is harmonic on  $W$ , or  $u(p) = +\infty$  for all  $p \in W$ .*

The notion of a subharmonic barrier (Section XV.4) is also local. The proof given in Section XV.4 shows that if  $W \cup \partial W$  is compact, and there is a subharmonic barrier at every point of  $\partial W$ , then the Dirichlet problem is solvable for  $W$ . Since there is a subharmonic barrier at every point on a boundary arc of  $W$ , we obtain the following.

**Theorem.** *Let  $W$  be a domain on a Riemann surface such that  $W \cup \partial W$  is compact and  $\partial W$  consists of a finite number of simple closed piecewise smooth boundary curves. Then every continuous function  $h(q)$  on  $\partial W$  is given by the boundary values of a (unique) harmonic function  $h(p)$  on  $W$ .*

As an application, we show that any Riemann surface can be exhausted by finite bordered subsurfaces.

**Theorem.** *For each compact subset  $K$  of a Riemann surface  $R$ , there is a domain  $W$  in  $R$  such that  $K \subset W$ ,  $W \cup \partial W$  is compact, and  $\partial W$  consists of a finite number of disjoint simple closed analytic curves.*

For the proof, we cover  $K$  by a finite number of coordinate disks, and then we connect these by finitely many more coordinate disks, to obtain a domain  $U_0 \subset R$  such that  $U_0 \supset K$  and  $U_0$  is a finite union of coordinate disks. It is straightforward then to construct a domain  $U \subset U_0$  such that  $U \supset K$  and  $\partial U$  consists of a finite number of smooth arcs. Let  $D_0$  be a small coordinate disk in  $U$ , with boundary  $\Gamma_0$  contained in  $U$ . We solve the Dirichlet problem on the domain  $U \setminus (D_0 \cup \Gamma_0)$  and obtain a harmonic function  $u$  with boundary values  $u(p) = 1$  on  $\Gamma_0$  and  $u(p) = 0$  on  $\partial U$ . Since the critical points of  $u$  are isolated points that can accumulate only



on  $\partial U \cup \Gamma_0$ , we can choose  $\varepsilon > 0$  small such that the level set  $\{u = \varepsilon\}$  does not contain a critical point of  $u$ , and further such that  $u > \varepsilon$  on  $K \cap U$ . Then the domain  $W = \{p \in U : u(p) > \varepsilon\} \cup \Gamma_0 \cup D_0$  has the desired properties.

### Exercises for XVI.2

1. Let  $w = w(z)$  be analytic on a domain  $D$  in the complex plane. Show that

$$\frac{\partial^2}{\partial z \partial \bar{z}} = \left| \frac{dw}{dz} \right|^2 \frac{\partial^2}{\partial w \partial \bar{w}}.$$

Deduce that a smooth function  $h(w)$  is harmonic on  $w(D)$  if and only if  $h(w(z))$  is harmonic on  $D$ .

2. Let  $W$  be an open subset of a Riemann surface such that  $W \cup \partial W$  is compact, and let  $u(p)$  be a subharmonic function on  $W$ . Show that if  $\limsup u(p) \leq c$  as  $p \rightarrow \partial W$ , then  $u(p) \leq c$  for all  $p \in W$ .
3. Let  $\{u_n\}$  be a sequence of positive harmonic functions on a Riemann surface  $R$  that converges uniformly on some coordinate disk. Show that the sequence converges uniformly on each compact subset of  $R$ .
4. Let  $\{u_n\}$  be a sequence of harmonic functions on a Riemann surface  $R$  that is uniformly bounded on each compact subset of  $R$  and that converges uniformly on some coordinate disk. Show that the sequence converges uniformly on each compact subset of  $R$ .
5. Show that any finite bordered Riemann surface can be obtained from a compact Riemann surface by excising a finite number of closed coordinate disks. *Hint.* Assume that there is one boundary curve in  $\partial R$ . By punching out a small disk  $D_0$  and solving a Dirichlet problem, find a positive harmonic function  $u$  on  $R \setminus D_0$  such that  $u = 0$  on  $\partial R$ . Use  $u$  and its harmonic conjugate to map  $\{0 < u < \varepsilon\}$  onto an annulus  $\{1 < |w| < \sigma\}$ . Take  $S = R \cup \{|w| \leq 1\}$ , and use  $\{|w| < \sigma\}$  as a coordinate disk.

### 3. Green's Function of a Surface

To define Green's function for a Riemann surface  $R$ , we adapt the definition given for an arbitrary planar domain. Fix a point  $q \in R$ , and let  $z(p)$  be a coordinate map for a coordinate disk at  $q$  satisfying  $z(q) = 0$ . Let  $\mathcal{F}_q$  be the family of subharmonic functions  $u(p)$  on  $R \setminus \{q\}$  such that  $u = 0$  off some compact subset of  $R$ , and  $u(p) + \log |z(p)|$  is subharmonic on the coordinate disk at  $q$ . Then  $\mathcal{F}_q$  is a Perron family of subharmonic functions on  $R \setminus \{q\}$ . The family is nonempty, since the constant function  $u = 0$  belongs to  $\mathcal{F}_q$ .

There are now two cases that can occur. If the upper envelope of  $\mathcal{F}_q$  is finite on  $R \setminus \{q\}$ , we say that **Green's function for  $R$  with pole at  $q$  exists**, and we denote it by

$$g(p, q) = \sup\{u(p) : u \in \mathcal{F}_q\}, \quad p \in R \setminus \{q\}.$$

Otherwise, the upper envelope of  $\mathcal{F}_q$  is  $+\infty$  at all points  $p \in R \setminus \{q\}$ , and in this case we say that **Green's function does not exist**.

**Theorem.** *Suppose Green's function  $g(p, q)$  with pole at  $q$  exists, and let  $z(p)$  be a coordinate map for a coordinate disk at  $q$  with  $z(q) = 0$ . Then  $g(p, q)$  is harmonic and (strictly) positive for  $p \in R \setminus \{q\}$ , and  $g(p, q) + \log |z(p)|$  is harmonic at  $q$ . If  $h(p)$  is a positive harmonic function on  $R \setminus \{q\}$  such that  $h(p) + \log |z(p)|$  is harmonic at  $q$ , then  $h(p) \geq g(p, q)$  for  $p \in R \setminus \{q\}$ .*

As the upper envelope of a Perron family, and further  $g(p, q)$  is harmonic, and  $g(p, q) \geq 0$ . Let  $\rho > 0$  be small, so that the coordinate disk  $\{|z(p)| \leq \rho\}$  is defined. The function defined to be  $-\log |z(p)| + \log \rho$  on this coordinate disk and 0 outside it belongs to  $\mathcal{F}_q$ . Consequently,

$$(3.1) \quad g(p, q) \geq -\log |z(p)| + \log \rho, \quad 0 < |z(p)| \leq \rho,$$

and  $g(p, q) \rightarrow +\infty$  as  $p \rightarrow q$ . Since  $g(p, q)$  cannot attain its minimum unless it is constant,  $g(p, q) > 0$  on  $R \setminus \{q\}$ . Let  $M$  be the maximum of  $g(p, q)$  on the coordinate circle  $\{|z(p)| = \rho\}$ . Any  $u \in \mathcal{F}_q$  satisfies  $u(p) \leq M$  for  $|z(p)| = \rho$ , and consequently,

$$u(p) + \log |z(p)| \leq M + \log \rho$$

for  $|z(p)| = \rho$ . Since  $u(p) + \log |z(p)|$  is subharmonic on the coordinate disk, this estimate persists for  $|z(p)| \leq \rho$ . Taking the supremum over  $u \in \mathcal{F}_q$ , we obtain

$$(3.2) \quad g(p, q) + \log |z(p)| \leq M + \log \rho, \quad |z(p)| \leq \rho.$$

Combining this with (3.1), we see that  $g(p, q) + \log |z(p)|$  is a bounded harmonic function on the punctured coordinate disk  $\{0 < |z(p)| < \rho\}$ . Since isolated singularities of bounded harmonic functions are removable,  $g(p, q) + \log |z(p)|$  is harmonic at  $q$ .

For the final statement of the theorem, suppose  $h(p)$  is a positive harmonic function on  $R \setminus \{q\}$  with a logarithmic pole at  $q$ . If  $u \in \mathcal{F}_q$ , then  $u - h$  is subharmonic on  $R$ , and  $u - h < 0$  off a compact subset of  $R$ . By the maximum principle,  $u - h < 0$  on  $R$ . Taking the upper envelope of such  $u$ , we obtain  $g(p, q) - h(p) \leq 0$ . This completes the proof.

Our next goal is to show that if  $g(p, q)$  exists for one point  $q \in R$ , then it exists for all  $q \in R$ . The proof is based on an auxiliary harmonic function and a clever estimation technique that merits special attention.

Fix  $q_0 \in R$ , and let  $z(p)$  be a coordinate map for a coordinate disk at  $q_0$  with  $z(q_0) = 0$ . Fix  $r > 0$  small so that the closed coordinate disk  $B_r = \{|z(p)| \leq r\}$  is defined. Let  $\mathcal{F}$  be the family of subharmonic functions  $u$  on  $R \setminus B_r$  such that  $u \leq 1$ , and  $u = 0$  off a compact subset of  $R$ . Since  $\mathcal{F}$  is a Perron family, the upper envelope

$$\omega(p) = \sup\{u(p) : u \in \mathcal{F}\}, \quad p \in R \setminus B_r,$$

is a harmonic function, which evidently satisfies  $0 \leq \omega \leq 1$ . The function  $\omega(p)$  can be viewed as the Perron solution to the Dirichlet problem on  $R \setminus B_r$  with boundary function 1 on  $\partial B_r$  and 0 at  $\infty$ . Since there is a subharmonic barrier at each point of  $\partial B_r$ ,  $\omega(p) \rightarrow 1$  as  $p \rightarrow \partial B_r$ . Now, there are two cases that can occur: Either  $0 < \omega < 1$  on  $R \setminus B_r$ , or  $\omega \equiv 1$  on  $R \setminus B_r$ .

**Lemma.** *If  $g(p, q)$  exists for some point  $q$  in the open coordinate disk  $B_r \setminus \partial B_r = \{|z(p)| < r\}$ , then  $0 < \omega < 1$  on  $R \setminus B_r$ . Conversely, if  $0 < \omega < 1$  on  $R \setminus B_r$ , then  $g(p, q)$  exists for all  $q \in B_r \setminus \partial B_r$ .*

The first statement is the easier to prove. Suppose  $g(p, q)$  exists, and choose  $c > 0$  such that  $g(p, q) \geq c$  for all  $p \in \partial B_r$ . By the maximum principle,  $u(p) \leq g(p, q)/c$  for all  $u$  in the Perron family  $\mathcal{F}$  defining  $\omega$ . Consequently,  $\omega(p) \leq g(p, q)/c$ . Now, the infimum of  $g(p, q)$  for  $p \in R$  is 0; if it were  $a > 0$ , then  $g(p, q) - a$  would be a positive harmonic function on  $R \setminus \{q\}$  with a logarithmic pole at  $q$ , contradicting the minimality property of  $g(p, q)$  in the preceding theorem. Hence the infimum of  $\omega(p)$  is also 0, and the case  $0 < \omega < 1$  obtains.

For the converse, suppose that  $0 < \omega < 1$  on  $R \setminus B_r$ . Choose  $s > r$  such that the closed coordinate disk  $B_s = \{|z(p)| \leq s\}$  is defined. Choose  $\kappa < 1$  such that  $\omega(p) \leq \kappa$  for  $p \in \partial B_s$ . Choose  $C > 0$  such that  $|\log |z(p) - z(q)|| \leq C$  for  $p \in \partial B_r$ . Let  $u \in \mathcal{F}_q$ , and let  $M$  be the maximum of  $u(p)$  for  $p \in \partial B_s$ . Since  $u(p) + \log |z(p) - z(q)|$  is subharmonic on  $B_s$  and bounded by  $M + C$  on  $\partial B_s$ , we have

$$u(p) + \log |z(p) - z(q)| \leq M + C, \quad p \in B_s.$$

In particular, this holds for  $p \in \partial B_r$ , and consequently,

$$u(p) \leq M + 2C, \quad p \in \partial B_r.$$

Since  $\omega(p) = 1$  on  $\partial B_r$ , we have

$$u(p) \leq (M + 2C)\omega(p)$$

for  $p \in \partial B_r$ , hence by the maximum principle for all  $p \in R \setminus B_r$ . If we take the maximum over  $p \in \partial B_s$ , we then obtain

$$M \leq (M + 2C)\kappa.$$

This can be solved for  $M$  to yield  $M \leq 2C\kappa/(1 - \kappa)$ . Thus we have shown that any  $u \in \mathcal{F}_q$  satisfies  $u(p) \leq 2C\kappa/(1 - \kappa)$  for  $p \in \partial B_s$ . Consequently, Green's function with pole at  $q$  exists.

The lemma shows that the set of  $q \in R$  for which Green's function  $g(p, q)$  exists is open, and the set of  $q \in R$  for which  $g(p, q)$  does not exist is also open. Since  $R$  is connected, one of these sets coincides with  $R$  and the other is empty. We have proved the following.

**Theorem.** *If Green's function with pole at  $q$  exists for some  $q \in R$ , then it exists for all  $q \in R$ .*

Another consequence of the lemma is the following theorem, which implies in particular that Green's function exists for any finite bordered Riemann surface with nonempty border.

**Theorem.** *Suppose the Riemann surface  $R$  is an open subset of a Riemann surface  $S$ , and suppose there is a boundary arc for  $R$  in  $S$ . Then Green's function for  $R$  exists.*

In this case, there is a subharmonic barrier at each point of the arc, so the auxiliary function  $\omega(p)$  tends to 0 as  $p \in R$  tends to the boundary arc. Hence we are in the case where  $0 < \omega < 1$ , and Green's function exists.

Next we prove a theorem of T. Radó to the effect that any Riemann surface is the union of an increasing sequence (!) of compact subsets. This property is useful, but it is not essential, in the sense that if we could not derive it from the definition of a Riemann surface, we would simply incorporate it into the definition as property (1.5). (In fact, this property is incorporated into the definition of complex manifolds of higher dimension, and it is incorporated into the definition of differentiable manifolds of all dimensions.)

**Theorem.** *Let  $R$  be a Riemann surface. There is an increasing sequence  $\{R_n\}$  of finite bordered subsurfaces of  $R$  such that  $\cup R_n = R$ .*

Suppose first that Green's function  $g(p, q)$  exists. Fix  $p_0 \neq q$ , and choose a sequence  $u_n \in \mathcal{F}_q$  such that  $u_n(p_0) \rightarrow g(p_0, q)$ . Let  $R_n$  be a finite bordered subsurface of  $R$  containing  $p_0$  and  $q$  such that  $u_n = 0$  off a compact subset of  $R_n$ . We can arrange also that  $R_n$  includes the compact set  $R_{n-1} \cup \partial R_{n-1}$ . We adjoin to  $R_n$  any component of  $R \setminus R_n$  that is compact, and we can assume that there are no compact components of  $R \setminus R_n$ . Now,  $g_n(p, q)$  is an increasing sequence, and  $u_n(p_0) \leq g_n(p_0, q) \leq g(p_0, q)$ . By Harnack's theorem,  $g_n(p, q)$  converges uniformly on compact subsets of  $R \setminus \{q\}$  to a harmonic function  $u(p)$  on  $R \setminus \{q\}$ . Since  $u(p) - g(p, q) \leq 0$  on  $R \setminus \{q\}$ , and  $u(p_0) - g(p_0, q) = 0$ , the strict maximum principle implies that  $u(p) - g(p, q) \equiv 0$ . Consequently,  $g_n(p, q)$  converges to  $g(p, q)$  uniformly on

compact subsets of  $\cup R_n$ , and in particular it converges uniformly on  $\partial R_n$  for all  $n$ .

We claim that  $\cup R_n = R$ . We argue by contradiction. Suppose there is a point  $p_0 \in R \setminus \cup R_n$ . Let  $D_0$  be a coordinate disk at  $p_0$  with coordinate map  $z(p)$  satisfying  $z(p_0) = 0$ . No boundary curve of  $\partial R_n$  lies completely in  $D_0$ , or else it would bound a compact component of  $R \setminus R_n$ . Hence by following a straight line to the nearest curve in  $\partial R_n$  and then following this to  $\partial D_0$ , we obtain a curve  $\gamma_n$  in  $D_0 \setminus R_n$  connecting  $p_0$  to  $\partial D_0$ . Let  $E$  be the lim sup of the  $\gamma_n$ 's, that is,  $p \in E$  if and only if there is a sequence  $p_n \in \gamma_n$  such that  $p_n \rightarrow p$ . Then  $E$  is a compact connected subset of  $R$ , and  $E$  is disjoint from  $R_k$  for each  $k$ , so  $E \subset R \setminus \cup R_n$ . Note that  $p_0 \in E$ , and since each  $\gamma_k$  meets  $\partial D_0$ , also  $E$  meets  $\partial D_0$ . Thus  $E$  has more than one point. By the condition given in Section XV.4, there is a subharmonic barrier at each point of  $E$  for the open set  $R \setminus E$ . Hence Green's function  $g_E(p, q)$  of the component of  $R \setminus E$  containing  $q$  tends to 0 as  $p \rightarrow E$ . Since  $g_n(p, q) \leq g_E(p, q)$  for all  $n$ , we then obtain  $g(p, q) \rightarrow 0$  as  $p \rightarrow E$ . However,  $g(p, q) > 0$  for all  $p$ , so that  $E$  is empty. This contradiction establishes the theorem in the case that Green's function exists.

In the case that Green's function does not exist, we excise a coordinate disk  $D_0$  and consider the surface  $R \setminus D_0$ . Green's function does exist for  $R \setminus D_0$ . We apply the result already proved to  $R \setminus D_0$ , and the theorem for  $R$  follows easily.

### Exercises for XVI.3

1. Show that if Green's function exists for  $S$ , and  $R$  is a subsurface of  $S$ , then Green's function exists for  $R$ , and  $g_R \leq g_S$ .
2. Show that if  $f : R \rightarrow S$  is an analytic map of  $R$  onto  $S$ , and if Green's function exists for  $S$ , then Green's function exists for  $R$ , and  $g_R(p, q) \leq g_S(f(p), f(q))$ .
3. Suppose Green's function  $g(p, q)$  exists for  $R$ . Let  $z(p)$  be a coordinate map at  $q$  with  $z(q) = 0$ . Show that if  $u(p)$  is a positive harmonic function on  $R \setminus \{q\}$  such that  $u(p) + \log |z(p)|$  is bounded below near  $q$ , then  $u(p) \geq g(p, q)$ .
4. Suppose Green's function  $g(p, q)$  exists for  $R$ . Let  $z(p)$  be a coordinate map at  $q$  with  $z(q) = 0$ . Show that if  $u(p)$  is a subharmonic function on  $R \setminus \{q\}$  such that  $u(p) = 0$  off some compact subset of  $R$ , and  $u(p) + \log |z(p)|$  is bounded above near  $q$ , then  $u(p) \leq g(p, q)$ .
5. Let  $R$  be a Riemann surface for which Green's function does not exist. Let  $K$  be a compact subset of  $R$ . Show that if  $u(p)$  is a subharmonic function on  $R \setminus K$  such that  $u(p)$  is bounded above and  $\limsup u(p) \leq 0$  as  $p \rightarrow \partial K$ , then  $u(p) \leq 0$  for all  $p \in R \setminus K$ .

*Remark.* If Green's function for  $R$  exists, and the function  $\omega$  is as in this section, then this statement fails for  $-\omega$ .

6. Show that Green's function for  $R$  exists if and only if there is a nonconstant subharmonic function on  $R$  that is bounded above.  
*Hint.* If Green's function  $g(p, q)$  exists, consider  $-g(p, q)$ . For the reverse implication, use the preceding exercise.

#### 4. Symmetry of Green's Function

We are interested now in how Green's function  $g(p, q)$  depends on the parameter  $q$ . Our goal is to prove the following.

**Theorem.** Suppose  $R$  is a Riemann surface for which Green's function  $g(p, q)$  exists. Then Green's function is symmetric,

$$g(p, q) = g(q, p), \quad p, q \in R, \quad p \neq q.$$

It suffices to prove the theorem in the case that  $R$  is a finite bordered Riemann surface. The general case can then be obtained by taking limits.

To separate the analytic difficulties from the topological difficulties, we first prove the theorem in the case that  $D$  is a bounded domain in the complex plane with analytic boundary. In this case, the proof is reminiscent of the proof of Pompeiu's formula. We excise two closed disks  $U_0, U_1$  of small radius  $\varepsilon > 0$  centered at points  $\zeta_0, \zeta_1 \in D$ . Let  $D_\varepsilon = D \setminus (U_0 \cup U_1)$ . The functions  $u_0(z) = g(z, \zeta_0)$  and  $u_1(z) = g(z, \zeta_1)$  are harmonic on  $D_\varepsilon \cup \partial D_\varepsilon$ . From Green's second formula we obtain

$$\int_{\partial D_\varepsilon} \left( u_0 \frac{\partial u_1}{\partial n} - u_1 \frac{\partial u_0}{\partial n} \right) ds = 0.$$

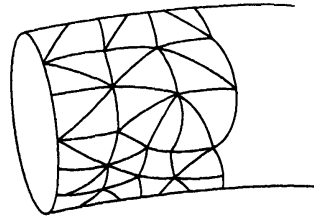
Since  $u_0 = u_1 = 0$  on  $\partial D$ , this becomes

$$\int_{\partial U_0} \left( u_0 \frac{\partial u_1}{\partial n} - u_1 \frac{\partial u_0}{\partial n} \right) ds + \int_{\partial U_1} \left( u_0 \frac{\partial u_1}{\partial n} - u_1 \frac{\partial u_0}{\partial n} \right) ds = 0.$$

We may reverse the direction of the unit normal vector, and take it to be the usual outer normal vectors to the disks  $U_0$  and  $U_1$ . With this switch in sign, the integral over  $\partial U_0$  is  $2\pi u_1(\zeta_0)$  and the integral over  $\partial U_1$  is  $-2\pi u_0(\zeta_1)$ , by Green's third formula (1.3). Hence  $u_1(\zeta_0) = u_0(\zeta_1)$ , and  $g(\zeta_0, \zeta_1) = g(\zeta_1, \zeta_0)$ .

If we had available the machinery of differential forms and Green's theorem for manifolds, we could carry over the proof for planar domains immediately to a finite bordered Riemann surface. However, we can also proceed directly by "triangulating"  $R$  and by applying Green's formula on each triangle.

Let  $R$  be a finite bordered Riemann surface, with Green's function  $g(p, q)$ , and fix points  $q_0, q_1 \in R$ . Our first step is to cut  $R$  into curvilinear "triangles"  $T_0, T_1, \dots, T_n$  such that each  $T_j$  is contained in some coordinate patch, say with coordinate map  $z_j(p)$ , and  $\partial T_j$  consists of three smooth arcs joining the three "vertices" of  $T_j$ . Thus if  $j \neq k$ , then either  $T_j$  is disjoint from  $T_k$ , or they share only a common vertex, or they share a common side consisting of a smooth arc that has opposite orientations with respect to the two triangles. We can carry out this triangulation procedure for any planar domain with piecewise smooth boundary, and since  $R$  is covered by a finite number of coordinate patches, we can triangulate  $R$  by focusing on the coordinate patches one by one. We assume that  $q_0$  lies inside  $T_0$  and  $q_1$  inside  $T_1$ .



Define  $u_0(p) = g(p, q_0)$  and  $u_1(p) = g(p, q_1)$ . From Green's third formula for  $j = 0, 1$ , and from Green's second formula for  $j \geq 2$ , we obtain

$$(4.1) \quad \int_{\partial T_j} \left( u_1 \frac{\partial u_2}{\partial n} - u_2 \frac{\partial u_1}{\partial n} \right) ds = \begin{cases} 2\pi u_1(q_0), & j = 0, \\ -2\pi u_0(q_1), & j = 1, \\ 0, & j \geq 2, \end{cases}$$

where the integral over  $\partial T_j$  is taken with respect to the coordinate map  $z_j(p)$ . Suppose  $\gamma$  is a common side of two triangles. Let  $v_0(p)$  and  $v_1(p)$  be conjugate harmonic functions for  $u_0(p)$  and  $u_1(p)$ , defined on  $\gamma$ . Then we can express the normal derivatives of the  $u_j$ 's in terms of the tangential derivatives of the  $v_j$ 's by the Cauchy-Riemann equations, and obtain

$$\begin{aligned} \int_{\gamma} \left( u_0 \frac{\partial u_1}{\partial n} - u_1 \frac{\partial u_0}{\partial n} \right) ds &= - \int_{\gamma} \left( u_0 \frac{\partial v_1}{\partial s} - u_1 \frac{\partial v_0}{\partial s} \right) ds \\ &= \int_{\gamma} u_1 dv_0 - u_0 dv_1. \end{aligned}$$

Now, the right-hand integral is independent of the coordinate map, and it changes sign if we reverse the direction of  $\gamma$ . Thus if we add up all the integrals in (4.1), the integrals over the common sides of adjacent triangles cancel, and we are left with the integrals over the sides of triangles on  $\partial R$ . Since both  $u_1$  and  $u_2$  are zero on  $\partial R$ , the integrals over these sides vanish, and we obtain

$$2\pi(u_0(q_1) - u_1(q_0)) = \sum \int_{\partial T_j} \left( u_1 \frac{\partial u_2}{\partial n} - u_2 \frac{\partial u_1}{\partial n} \right) ds = 0.$$

Thus  $u_0(q_1) = u_1(q_0)$ , and the symmetry is established.

**Example.** Green's function  $g(z, \zeta) = \log |(z - \zeta)/(1 - \bar{\zeta}z)|$  for the open unit disk  $\mathbb{D}$  is symmetric in  $z$  and  $\zeta$ .

#### Exercises for XVI.4

1. Let  $R$  be a finite bordered Riemann surface with border  $\partial R$ , and fix  $q \in R$ . Let  $h(p)$  be the (locally defined) conjugate harmonic function for Green's function  $g(p, q)$  for  $R$ . Show that if  $u(p)$  is a harmonic function on  $R$  that is smooth up to  $\partial R$ , then

$$u(q) = \frac{1}{2\pi} \int_{\partial R} u(p) dh(p).$$

Show further that  $h(p)$  is strictly increasing on  $\partial R$ , and the total increase of  $h(p)$  around the curves in  $\partial R$  is  $2\pi$ . *Remark.* The integral can be defined by breaking  $\partial R$  into arcs contained in coordinate disks. It does not depend on the coordinate maps, but it does depend on the orientation of the curves in  $\partial R$ , which must be oriented positively with respect to  $R$ .

### 5. Bipolar Green's Function

Not every Riemann surface has a Green's function. However, every surface *does* have a bipolar Green's function. In this section we will prove the existence of bipolar Green's function, and in the next we will use it to prove the celebrated uniformization theorem.

Let  $q_1, q_2$  be distinct points of the Riemann surface  $R$ , and let  $\Delta_1$  and  $\Delta_2$  be disjoint coordinate disks containing  $q_1$  and  $q_2$ , respectively, with coordinate maps  $z_1(p)$  and  $z_2(p)$  satisfying  $z_1(q_1) = 0$  and  $z_2(q_2) = 0$ . We define **bipolar Green's function** with poles at  $q_1$  and  $q_2$  to be any harmonic function  $G(p, q_1, q_2)$  on  $R \setminus \{q_1, q_2\}$  such that

$$(5.1) \quad G(p, q_1, q_2) + \log |z_1(q)| \text{ is harmonic at } q_1,$$

$$(5.2) \quad G(p, q_1, q_2) - \log |z_2(q)| \text{ is harmonic at } q_2,$$

$$(5.3) \quad G(p, q_1, q_2) \text{ is bounded on } R \setminus (\Delta_1 \cup \Delta_2).$$

Thus  $G(p, q_1, q_2)$  has a logarithmic pole at  $q_1$ , and  $-G(p, q_1, q_2)$  has a logarithmic pole at  $q_2$ . Note that  $G(p, q_1, q_2)$  is not uniquely determined. It is unique up to adding a bounded harmonic function.

**Example.** Bipolar Green's function for the Riemann sphere  $\mathbb{C}^*$  with poles at  $q_1 = 0$  and  $q_2 = \infty$  is given by  $G(z, 0, \infty) = -\log |z|$ . In this case,  $G(z, 0, \infty)$  is uniquely determined up to adding a constant, since a bounded harmonic function on  $\mathbb{C}^*$  is constant.



**Example.** If Green's function  $g(p, q)$  exists for the Riemann surface  $R$ , then

$$G(p, q_1, q_2) = g(p, q_1) - g(p, q_2)$$

is a bipolar Green's function for  $R$ .

**Theorem.** For each pair of distinct points  $q_1, q_2$  on a Riemann surface, there is a bipolar Green's function  $G(p, q_1, q_2)$ .

The technical part of the proof is contained in the following lemma.

**Lemma.** Let  $S$  be a finite bordered Riemann surface, and let  $q_1, q_2$  be distinct points of  $S$ . Let  $B_1 = \{|z_1(p)| \leq \sigma\}$  and  $B_2 = \{|z_2(p)| \leq \sigma\}$  be disjoint closed coordinate disks, where  $z_1(q_1) = 0, z_2(q_2) = 0$ . There is a constant  $C > 0$  such that

$$|g_R(p, q_1) - g_R(p, q_2)| \leq C, \quad p \in R \setminus (B_1 \cup B_2),$$

for all Riemann surfaces  $R$  containing  $S \cup \partial S$  for which Green's function  $g_R(p, q)$  exists.

It suffices to establish the lemma for finite bordered Riemann surfaces  $R$  containing  $S \cup \partial S$ . Let  $\rho > 0$  satisfy  $\rho < \sigma$ . For  $j = 1, 2$ , let  $A_j$  be the closed coordinate disk  $\{|z_j(p)| \leq \rho\}$ , and let  $M_j = M_j(R)$  be the maximum of  $g_R(p, q_j)$  on  $\partial A_j$ . For  $p \in \partial B_j$  we have

$$g_R(p, q_j) + \log |z_j(p)| \leq \max\{g_R(q, q_j) : q \in \partial B_j\} + \log \sigma,$$

and  $g_R(p, q_j) + \log |z_j(p)|$  is harmonic on  $B_j$ . By the maximum principle, the estimate persists for all  $p \in B_j$ . Taking the supremum over  $p \in \partial A_j$ , we obtain

$$M_j + \log \rho \leq \max\{g_R(q, q_j) : q \in \partial B_j\} + \log \sigma.$$

Thus there is  $p_j \in \partial B_j$  such that  $M_j + \log \rho \leq g_R(p_j, q_j) + \log \sigma$ , or

$$M_j - g_R(p_j, q_j) \leq \log(\sigma/\rho).$$

Now,  $M_j - g_R(p, q_j)$  is a positive harmonic function on  $S \setminus (A_1 \cup A_2)$ . If we apply the Harnack estimate from Section 2 to the Riemann surface  $S \setminus (A_1 \cup A_2)$  and the compact subset  $\partial B_1 \cup \partial B_2$ , we obtain a constant  $C_0$ , independent of  $R$ , such that  $M_j - g_R(p, q_j) \leq C_0$  for  $p \in \partial B_1 \cup \partial B_2$ . Thus

$$(5.4) \quad M_j - C_0 \leq g_R(p, q_j) \leq M_j, \quad p \in \partial B_1 \cup \partial B_2.$$

Since  $g_R(p, q_1)$  is harmonic for  $p \in B_2$  and satisfies (5.4) on  $\partial B_2$ , it must satisfy (5.4) for all  $p \in B_2$ . In particular,

$$M_1 - C_0 \leq g_R(q_2, q_1) \leq M_1.$$

Similarly,

$$M_2 - C_0 \leq g_R(q_1, q_2) \leq M_2.$$

Since  $g_R(q_2, q_1) = g_R(q_1, q_2)$ , these estimates imply that  $|M_1 - M_2| \leq C_0$ . From (5.4) we then obtain

$$|g_R(p, q_1) - g_R(p, q_2)| \leq 2C_0, \quad p \in \partial B_1 \cup \partial B_2.$$

Since Green's function vanishes on  $\partial R$ , this estimate persists on  $R \setminus (B_1 \cup B_2)$ , by the maximum principle. This establishes the lemma, with  $C = 2C_0$ .

Now we complete the proof by appealing to compactness results for uniformly bounded families of harmonic functions. Conceptually, the easiest way to proceed is to use Radó's theorem and exhaust  $R$  by a sequence of finite bordered Riemann surfaces with Green's functions  $g_n(p, q)$ . The differences  $G_n(p, q_1, q_2) = g_n(p, q_1) - g_n(p, q_2)$  are then uniformly bounded for  $p$  outside of coordinate disks containing  $q_1$  and  $q_2$ . We pass to a convergent subsequence, and the limit is bipolar Green's function  $G(p, q_1, q_2)$ .

There is a useful trick that allows us to bypass Radó's theorem. It is to approximate  $R$  by surfaces  $R \setminus B_\varepsilon$ , where  $B_\varepsilon$  is a closed coordinate disk  $\{|z_0(p)| \leq \varepsilon\}$  centered at some arbitrary point  $p_0 \in R$ . Green's function  $g_\varepsilon(p, q)$  exists for  $R \setminus B_\varepsilon$ , and we can form bipolar Green's function  $g_\varepsilon(p, q_1) - g_\varepsilon(p, q_2)$ . The lemma and compactness results allow us to pass to a normal limit through some sequence  $\varepsilon_j \rightarrow 0$ , to obtain a bipolar Green's function  $G(p, q_1, q_2)$  for  $R \setminus \{p_0\}$ . Since  $G(p, q_1, q_2)$  is bounded near the isolated singularity  $p_0$ , it extends harmonically to  $p_0$  and is a bipolar Green's function for  $R$ .

### Exercises for XVI.5

1. Determine all bipolar Green's functions for the complex plane  $\mathbb{C}$ .
2. Determine all bipolar Green's functions for the Riemann sphere  $\mathbb{C}^*$ .
3. Write out the details for the proof that bypasses Radó's theorem.  
*Hint.* See Exercise 2.3.

## 6. The Uniformization Theorem

A Riemann surface  $R$  is **simply connected** if any closed path in  $R$  can be deformed continuously to a point. This occurs if and only if any two paths in  $R$  from  $p_0$  to  $p_1$  can be deformed to each other through a continuous family of paths from  $p_0$  to  $p_1$ . If  $R$  is simply connected, and if  $S$  is a Riemann surface conformally equivalent to  $R$ , then  $S$  is simply connected. Our aim in this section is to prove the following fundamental theorem of Koebe and Poincaré.

**Theorem (Uniformization Theorem for Riemann Surfaces).** *Each simply connected Riemann surface is conformally equivalent to either the open unit disk  $\mathbb{D}$ , the complex plane  $\mathbb{C}$ , or the Riemann sphere  $\mathbb{C}^*$ .*

One of the main ingredients of the proof is the idea of analytic continuation along a path and the monodromy theorem. Since analytic continuation along paths is a local notion, it can be extended to paths on Riemann surfaces. The proof of the monodromy theorem (Section V.8) is local, and so the monodromy theorem holds on Riemann surfaces.

Our strategy is to split the proof into two cases. First we show that if Green's function for  $R$  exists, it can be used to map  $R$  conformally to the open unit disk. Then if Green's function for  $R$  does not exist, we use bipolar Green's function to map  $R$  to the sphere or the punctured plane.

So suppose  $R$  is a simply connected Riemann surface for which Green's function exists. Fix a point  $q_0 \in R$ . Since  $g(p, q_0)$  has a logarithmic pole at  $q_0$ , there is an analytic function  $\varphi$  defined near  $q_0$  with a simple zero at  $q_0$  such that  $|\varphi(p)| = e^{-g(p, q_0)}$ . The function  $\varphi$  can be continued analytically along any path in  $R$  from  $q_0$  to any other point  $p$ , by continuing the harmonic conjugate function of  $g(p, q_0)$  along the path and taking an exponential. Since  $R$  is simply connected, the monodromy theorem asserts that the analytic continuation does not depend on the path from  $q_0$  to  $p$ , and consequently, it defines an analytic function  $\varphi$  on  $R$  such that

$$|\varphi(p)| = e^{-g(p, q_0)}, \quad p \in R.$$

In particular,  $|\varphi(p)| < 1$  for  $p \in R$ , and  $\varphi$  has only one zero, a simple zero at  $q_0$ . Fix  $q_1 \in R$ , and define

$$\psi(p) = \frac{\varphi(p) - \varphi(q_1)}{1 - \overline{\varphi(q_1)}\varphi(p)}, \quad p \in R.$$

Then  $\psi$  is analytic on  $R$ ,  $|\psi(p)| < 1$  for  $p \in R$ , and  $\psi(q_1) = 0$ . If  $u \in \mathcal{F}_{q_1}$ , then  $u(p) + \log |\psi(p)|$  is subharmonic on  $R$ , including at  $q_1$ , since  $\psi(q_1) = 0$ , and  $u(p) + \log |\psi(p)| < 0$  off a compact subset of  $R$ . By the maximum principle,  $u(p) + \log |\psi(p)| < 0$  on  $R$ . Taking the supremum over  $u \in \mathcal{F}_{q_1}$ , we obtain

$$(6.1) \quad g(p, q_1) + \log |\psi(p)| \leq 0, \quad p \in R.$$

Since  $\psi(q_0) = -\varphi(q_1)$ , we obtain using the symmetry of Green's function that

$$g(q_0, q_1) + \log |\psi(q_0)| = g(q_0, q_1) - g(q_1, q_0) = 0.$$

By the strict maximum principle, equality holds in (6.1). Thus  $\log |\psi(p)| = -g(p, q_1)$  for all  $p \in R$ , and consequently,  $\psi$  has no zeros on  $R \setminus \{q_1\}$ . It follows that  $\varphi$  assumes the value  $\varphi(q_1)$  only at  $p = q_1$ . Since  $q_1$  is arbitrary,  $\varphi$  is one-to-one. Thus  $\varphi$  maps  $R$  conformally onto a domain  $\varphi(R)$  in the unit disk, and further  $\varphi(R)$  is simply connected. By the Riemann mapping

theorem,  $\varphi(R)$  is conformally equivalent to the open unit disk, and then so is  $R$ .

For the remaining case of the proof, we require the following lemma. (For a more general statement, see the exercises for Section 3.)

**Lemma.** *If there is a nonconstant bounded analytic function on a Riemann surface  $R$ , then Green's function for  $R$  exists.*

In this case, for a prescribed point  $q \in R$  we can find a nonconstant analytic function  $f$  on  $R$  such that  $f(q) = 0$  and  $|f(p)| < 1$  for  $p \in R$ . If  $u \in \mathcal{F}_q$ , then  $u(p) + \log |f(p)|$  is subharmonic on  $R$ , and  $u(p) + \log |f(p)| < 0$  off a compact subset of  $R$ . By the maximum principle,  $u(p) + \log |f(p)| < 0$  on  $R$ . This provides an upper bound  $-\log |f(p)|$  for the functions in  $\mathcal{F}_q$ , and consequently, Green's function with pole at  $q$  exists.

Now suppose that Green's function for  $R$  does not exist. Let  $G(p, q_1, q_2)$  be a bipolar Green's function for  $R$ . Since  $R$  is simply connected, we can then find by analytic continuation a meromorphic function  $\varphi$  on  $R$  such that

$$|\varphi(p)| = e^{-G(p, q_1, q_2)}, \quad p \in R.$$

In particular,  $\varphi$  has only one zero on  $R$ , a simple zero at  $q_1$ , and only one pole on  $R$ , a simple pole at  $q_2$ . Further, there are coordinate disks  $B_1$  at  $q_1$  and  $B_2$  at  $q_2$  such that

$$\frac{1}{C} \leq |\varphi(p)| \leq C, \quad p \in R \setminus (B_1 \cup B_2).$$

We claim that  $\varphi$  is one-to-one. To see this, let  $q_0$  be any point of  $R$  distinct from  $q_1$  and  $q_2$ , select a bipolar Green's function  $G(p, q_0, q_2)$ , and let  $\varphi_0$  be a meromorphic function such that

$$|\varphi_0(p)| = e^{-G(p, q_0, q_2)}, \quad p \in R.$$

Consider the quotient

$$\psi(p) = \frac{\varphi(p) - \varphi(q_0)}{\varphi_0(p)}, \quad p \in R.$$

The poles at  $q_2$  cancel, so  $\psi$  is a bounded analytic function on  $R$ . Since Green's function does not exist, the lemma shows that  $\psi$  is constant. Since the poles at  $q_2$  are both simple, the quotient  $\psi$  is not zero at  $q_2$ , and consequently,  $\psi(p) \equiv c \neq 0$ . It follows that  $\varphi$  assumes the value  $\varphi(q_0)$  at only one point, at  $p = q_0$ , and consequently,  $\varphi$  is one-to-one. Now,  $\varphi$  maps  $R$  conformally onto a simply connected domain  $\varphi(R)$  on the Riemann sphere  $\mathbb{C}^*$ . If  $\mathbb{C}^* \setminus \varphi(R)$  has more than one point, then by the Riemann mapping theorem  $\varphi(R)$  can be mapped conformally onto the unit disk, and consequently, Green's functions for  $\varphi(R)$  and hence for  $R$  exist. We conclude that  $\mathbb{C}^* \setminus \varphi(R)$  has at most one point. Thus either  $\varphi(R) = \mathbb{C}^*$ , or else  $\varphi(R)$  is the punctured sphere, which is conformally equivalent to  $\mathbb{C}$ .

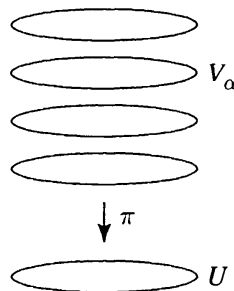
## Exercises for XVI.6

1. Show that the symmetry of Green's function used in the proof can be derived directly by obtaining an inequality from (6.1) and then interchanging the roles of  $q_0$  and  $q_1$ .
2. Consider the fundamental domain  $U$  for the modular group  $\mathcal{U}$  defined in Exercise 1.14.
  - (a) Show that with appropriate identification of points of  $\partial U$  the fundamental domain can be made into a Riemann surface.
  - (b) Show that the Riemann surface is conformally equivalent to the complex plane.
  - (c) Show that there is an analytic function  $J(\tau)$  on the upper half-plane  $\mathbb{H}$  such that  $J(\tau) = J(\tau')$  if and only if there is  $f \in \mathcal{U}$  such that  $f(\tau) = \tau'$ . This occurs if and only if the Riemann surface  $T_\tau$  (defined in Exercises 1.13) is conformally equivalent to  $T_{\tau'}$ .
  - (d) Show that  $J(\tau)$  has critical points at  $(\sqrt{3}i \pm 1)/2$  of order 2 and at  $i$  of order 1.

## 7. Covering Surfaces

We have seen how a torus is obtained from the complex plane by identifying points that are congruent modulo a lattice  $L$ . In this section we indicate how any Riemann surface can be obtained from a simply connected Riemann surface in roughly the same manner, by identifying congruent points. The main new ideas are “covering maps” and “covering transformations.”

Let  $R$  and  $S$  be two Riemann surfaces. An analytic map  $\pi : S \rightarrow R$  is a **covering map** if each  $p \in R$  belongs to an open subset  $U$  of  $R$  whose inverse image  $\pi^{-1}(U) = \cup V_\alpha$  is a union of disjoint open subsets  $V_\alpha$  of  $S$ , each of which is mapped one-to-one by  $\pi$  onto  $U$ . If  $U$  is a coordinate disk in  $R$  with coordinate map  $z(p)$ , then each  $V_\alpha$  is a coordinate disk in  $S$  with coordinate map  $(z \circ \pi)(q)$ .

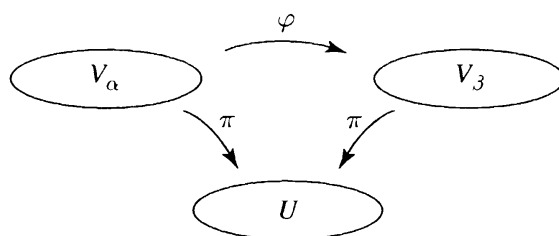


**Example.** Let  $S$  be the Riemann surface of  $\sqrt{z}$ , constructed from two copies of the slit plane  $\mathbb{C} \setminus (-\infty, 0]$  by identifying edges appropriately. The natural map  $\pi$  of  $S$  onto the punctured plane  $\mathbb{C} \setminus \{0\}$  is a covering map.

If  $D_0$  is a disk in  $\mathbb{C} \setminus (-\infty, 0]$ , then  $\pi^{-1}(D_0)$  consists of two disjoint disks, one on each sheet. If  $D_0$  is a disk in  $\mathbb{C} \setminus \{0\}$  overlapping the negative axis  $(-\infty, 0)$ , then  $\pi^{-1}(D_0)$  still consists of two disjoint coordinate disks, one in each of the coordinate patches  $U_3$  and  $U_4$  described in Section 1.

**Example.** There is a natural projection  $\pi$  of the Riemann surface of  $\log z$  onto the punctured complex plane  $\mathbb{C} \setminus \{0\}$ . In this case,  $\pi$  is a covering map, and each  $\pi^{-1}(D_0)$  consists of infinitely many coordinate disks on the surface, one for each sheet used in constructing the surface.

**Example.** Let  $T$  be the torus determined by the lattice  $L$  generated by  $\omega_1$  and  $\omega_2$ . The natural projection  $\pi(z) = z + L$ ,  $z \in \mathbb{C}$ , is then a covering map of  $\mathbb{C}$  onto  $T$ . If  $D_\varepsilon$  is a small open disk in  $\mathbb{C}$ , then  $D_\varepsilon + L$  is a coordinate disk in the torus, and  $\pi^{-1}(D_\varepsilon + L)$  consists of all translates  $D_\varepsilon + m\omega_1 + n\omega_2$ , where  $m$  and  $n$  are integers. Each of these translates is an open disk that is mapped one-to-one onto the coordinate disk in  $T$ .



Suppose  $\pi : S \rightarrow R$  is a covering map. A **covering transformation** is a one-to-one analytic map  $\varphi$  of  $S$  onto  $S$  such that  $\pi(\varphi(p)) = \pi(p)$  for all  $p \in S$ . The composition of two covering transformations is a covering transformation and the inverse of a covering transformation is a covering transformation. A covering transformation  $\varphi$  permutes the points of each set  $\pi^{-1}(\{q\})$ ,  $q \in R$ . Further, if  $U$  is a coordinate disk on  $R$  such that  $\pi^{-1}(U) = \cup V_\alpha$  is a disjoint union of coordinate disks on  $S$  as above, then  $\varphi$  permutes the  $V_\alpha$ 's, and  $\varphi$  maps each  $V_\alpha$  one-to-one onto some  $V_\beta$  by  $\varphi = \pi_\beta^{-1} \circ \pi_\alpha$  on  $V_\alpha$ , where  $\pi_\alpha$  is the restriction of  $\pi$  to  $V_\alpha$ . In particular, if  $\varphi(p) = p$  for some point  $p \in V_\alpha$ , then  $\varphi = \pi_\alpha^{-1} \circ \pi_\alpha$  is the identity map on the entire coordinate disk  $V_\alpha$ . Since  $S$  is connected, the uniqueness principle holds for analytic maps of  $S$  (see the exercises in Section 1), and consequently,  $\varphi$  is the identity map of  $S$ . We state this result formally.

**Theorem.** Let  $\pi : S \rightarrow R$  be a covering map, and let  $\varphi : S \rightarrow S$  be a covering transformation. Then either  $\varphi$  is the identity transformation  $\varphi(p) \equiv p$ , or  $\varphi$  has no fixed points.

**Example.** Consider the covering map  $\pi : \mathbb{C} \rightarrow T$  of the complex plane onto the torus determined by the lattice  $L$  generated by  $\omega_1$  and  $\omega_2$ . Any